Electronic supplementary material - Hydrogeology Journal

Groundwater conceptualization and modeling using distributed SWAT-based recharge for the semi-arid agricultural Neishaboor plain, Iran

A. Izady*[1, 5], K. Davary[1], A. Alizadeh[1], A. N. Ziaei[1], S. Akhavan[2], A. Alipoor[3], A. Joodavi[4], M. L. Brusseau[5]

[1]*Water Engineering Dept., College of Agriculture, Ferdowsi University of Mashhad, Mashhad, Iran*
[2]*Water Engineering Dept., College of Agriculture, Bu-Ali Sina University, Hamadan, Iran*
[3]*Neishaboor Water Authority, Neishaboor, Iran*
[4]*Earth Sciences Dept., College of Sciences, Shiraz University, Shiraz, Iran*
[5]*Soil, Water and Environmental Science Department, University of Arizona, Tucson, Arizona, USA*

**Corresponding author email: az.izady@gmail.com*

The following supplementary explanations have been included for more clarification of some sections of the main article.

1. Groundwater conceptual model

1.1. The bedrock

To determine the position of the bedrock, the results of a mapping effort using geo-electric sounding (in 21 cross-sections, with 232 sounding points) was used (Ministry of Water and Electricity, 1966). The distribution of these geo-electric soundings was deemed to provide a satisfactory resolution of bedrock topology. To achieve maximum possible clarity for the bedrock, a combination of old and new information was considered. New information consisted of withdrawal-well logs, scattered geo-electric soundings (49 for other applications), and new borehole logs. Besides, local experts and/or drillers were consulted in many cases to resolve conflicts or uncertainties. Of 1638 well logs, 329 were located on the geo-electric cross-sections or adjacent to those. Out of these 329 wells, only 137 were drilled to the bedrock. In addition, valuable information extracted from five exploratory wells was used (Fig.

S1). Data were compared for these five wells and the quality of the data was also evaluated with reference to other useful information, such as depth and discharge of the adjacent withdrawal wells, regional geological formation, and local expert knowledge. Then, all verified data were assigned to the geo-electric cross-sections to determine the depth of bedrock at each sounding point (Fig. S2 and S3).

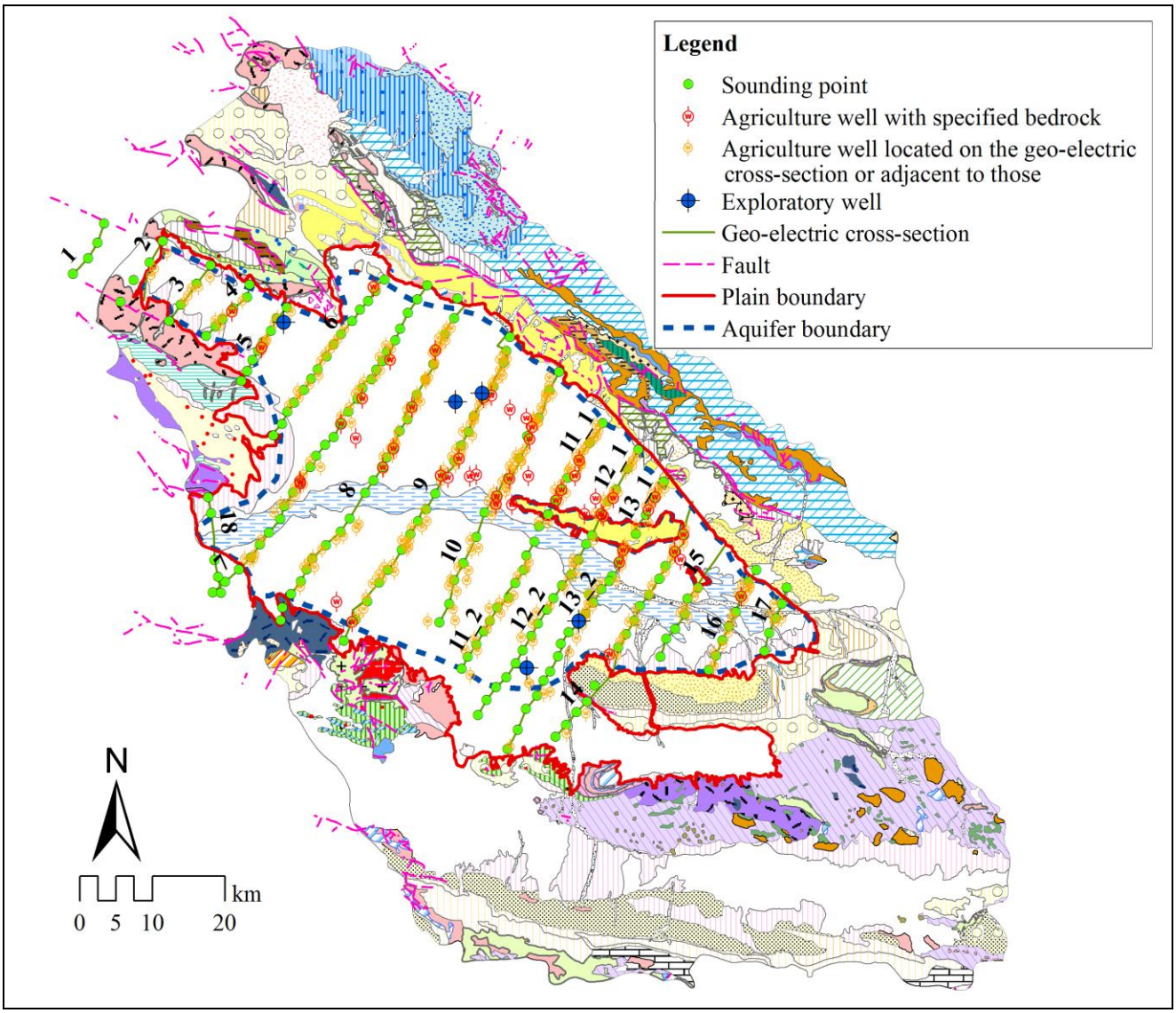


Fig. S1 Plan view of the geo-electric cross sections along with agriculture wells adjacent to them, exploratory wells and faults. See main article for description of geology

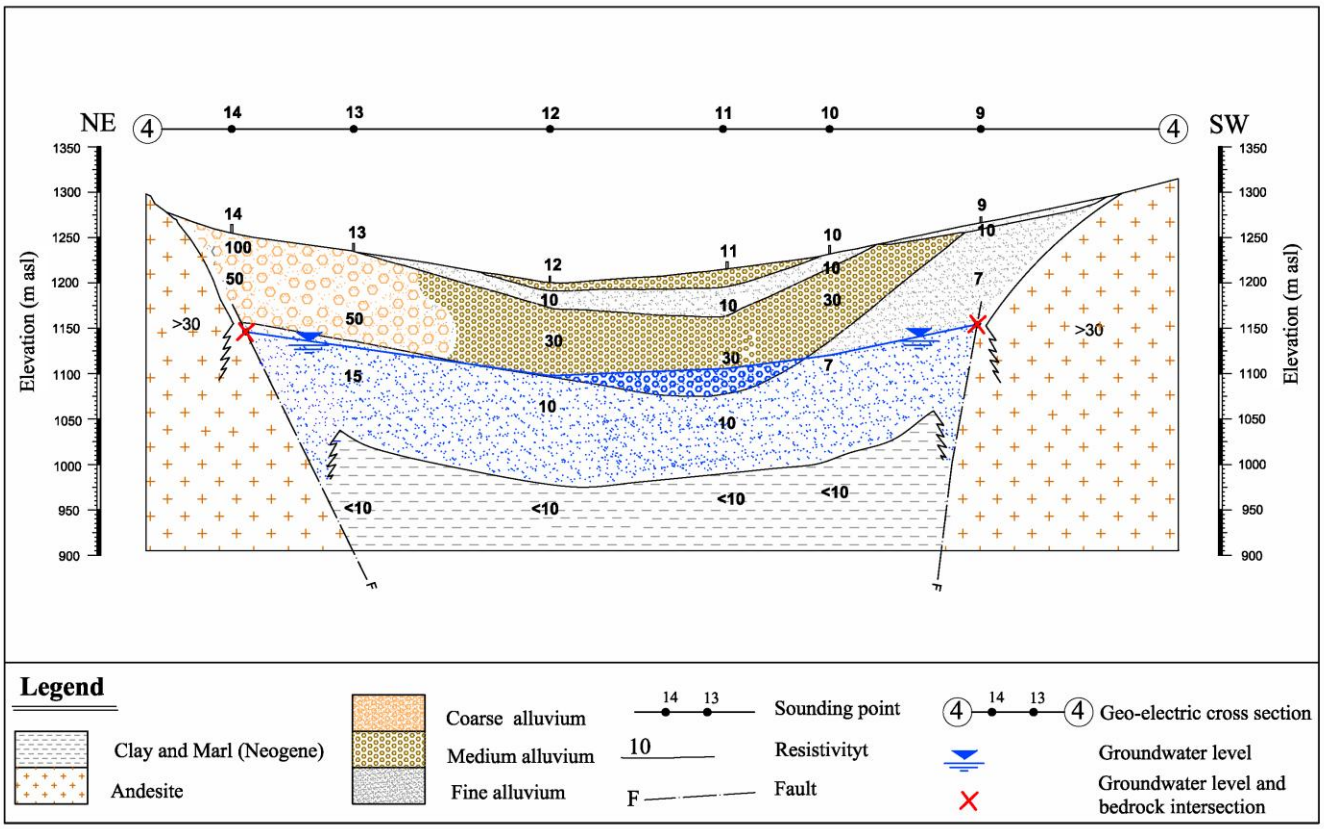


Fig. S2 The geo-electric cross section # 4 (m asl is meters above sea level). (see Fig. S1 for location)

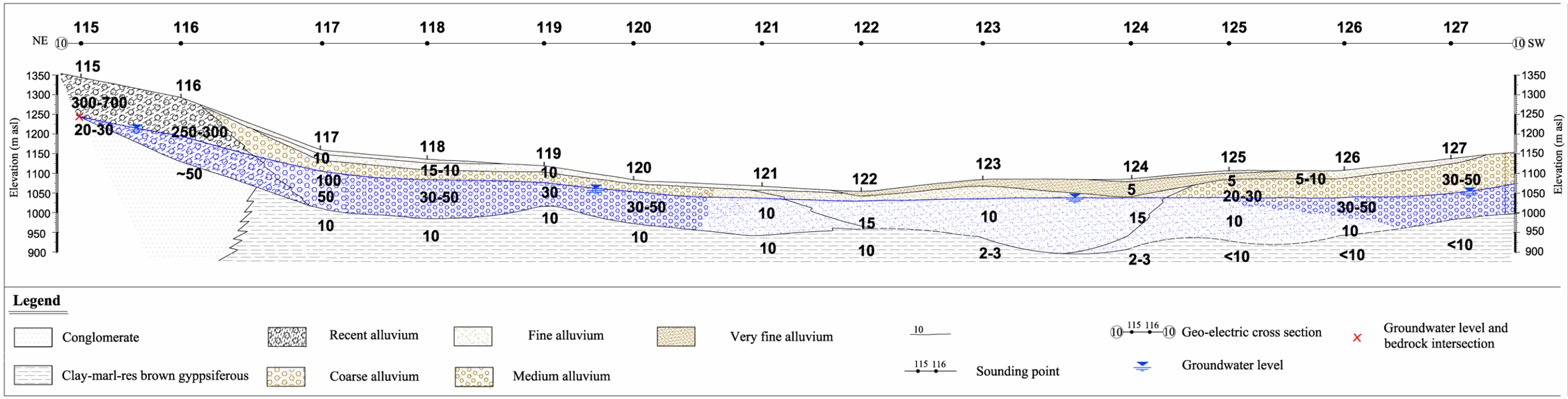


Fig. S3 The geo-electric cross section # 10 (m asl is meters above sea level). (see Fig. S1 for location)

After determining depth of bedrock for all cross-sections, the bedrock position became almost clear at 320 points within the plain (Fig. S4). The bedrock depth varies from 0 to 260 m in the different parts of the plain. A Neogene and marl rock outcrop resides in the east of the plain, which was considered as an impermeable barrier to the groundwater flow system. The alluvial deposits in the north of the plain formed the deepest part of the bedrock, ranging from 180 to 260 m. This thick alluvial deposit near to the mountains is filled with Quaternary and upper Pliocene sediments with great groundwater storage capacity. Also, this region catches mountains runoff via alluvial fans which is a main source of groundwater recharge.

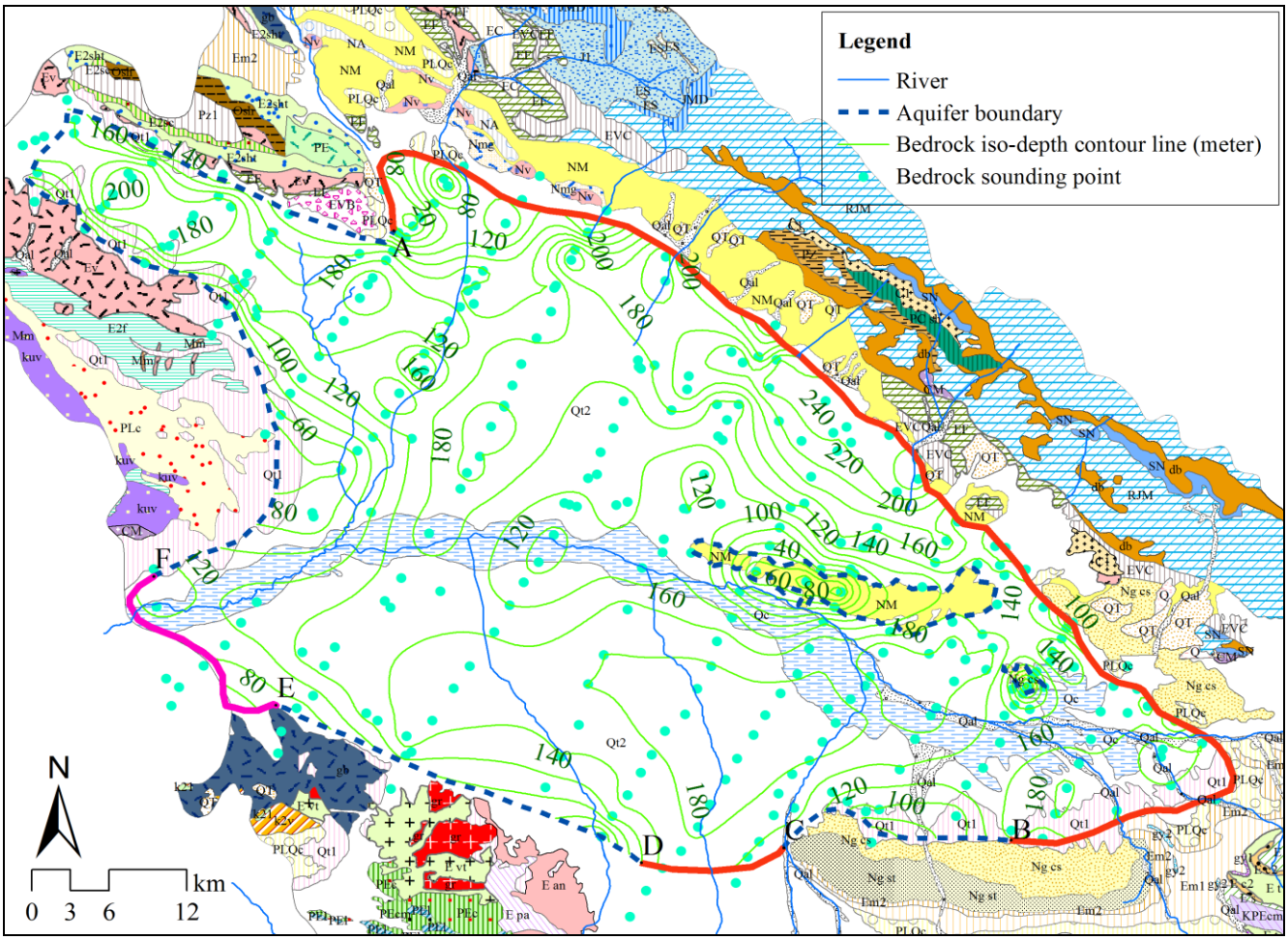


Fig. S4 Bedrock-depth contour lines for the Neishaboor aquifer, spatial distribution of sounding points and boundary conditions (the red lines (A-B and C-D) have inflow boundary conditions, and the pink line (E-F) has outflow boundary conditions)

1.2. Hydrodynamic properties

The spatially distributed hydrodynamics properties were estimated using different methods. In the 1969 report, the specific yield was estimated using a granulation method at 45 points. In this method, soil samples were collected from different depths and their soil textures were determined. The specific yield was approximated for each soil texture (see Table S1). The saturation thickness of each soil type was determined and the weighted average method was used to calculate the specific yield for each well. The observed decline in groundwater level (from 1969 to 2012) was accounted for by modification of the reported specific yield values. This was done by determining the texture of the remaining saturated media at depth using drilling logs of the original or adjacent wells.

The hydraulic conductivity was calculated from the estimated specific yield using relationships reported in the literature. Two different experimental equations were used. The first is:

$$Sy = \sqrt{K} \tag{S1}$$

where *S*y is specific yield (%) and *K* is hydraulic conductivity (cm/day). The second equation (Ahuja et al.1989) is given as:

$$K = 0.2561e^{0.231 \times Sy} \tag{S2}$$

Where, again, *S*y is specific yield (%) but *K* is hydraulic conductivity (cm/hr).

Table S1 Specific yield for different soil textures and materials (after Johnson 1967)

Material	Specific yield (%)		
	Maximum	Minimum	Average
Clay	5	0	2
Sandy clay	12	3	7
Silt	19	3	18
Fine sand	28	10	21
Medium sand	32	15	26
Coarse sand	35	20	27
Gravelly sand	35	20	25
Fine gravel	35	21	25
Medium gravel	26	13	23
Coarse gravel	26	12	22

Also, some point measures of the specific yield and hydraulic conductivity have been reported in the literature for this region for particular purposes (seven pumping wells, dam construction, and steel manufacturer). Estimated hydraulic conductivity values – ranging from 1.5 to 30 m/day - show that east, south and southwest parts of the plain have larger values. However, near to the plain's outlet, hydraulic conductivity decreases due to fine grained alluviums (Fig. S5).

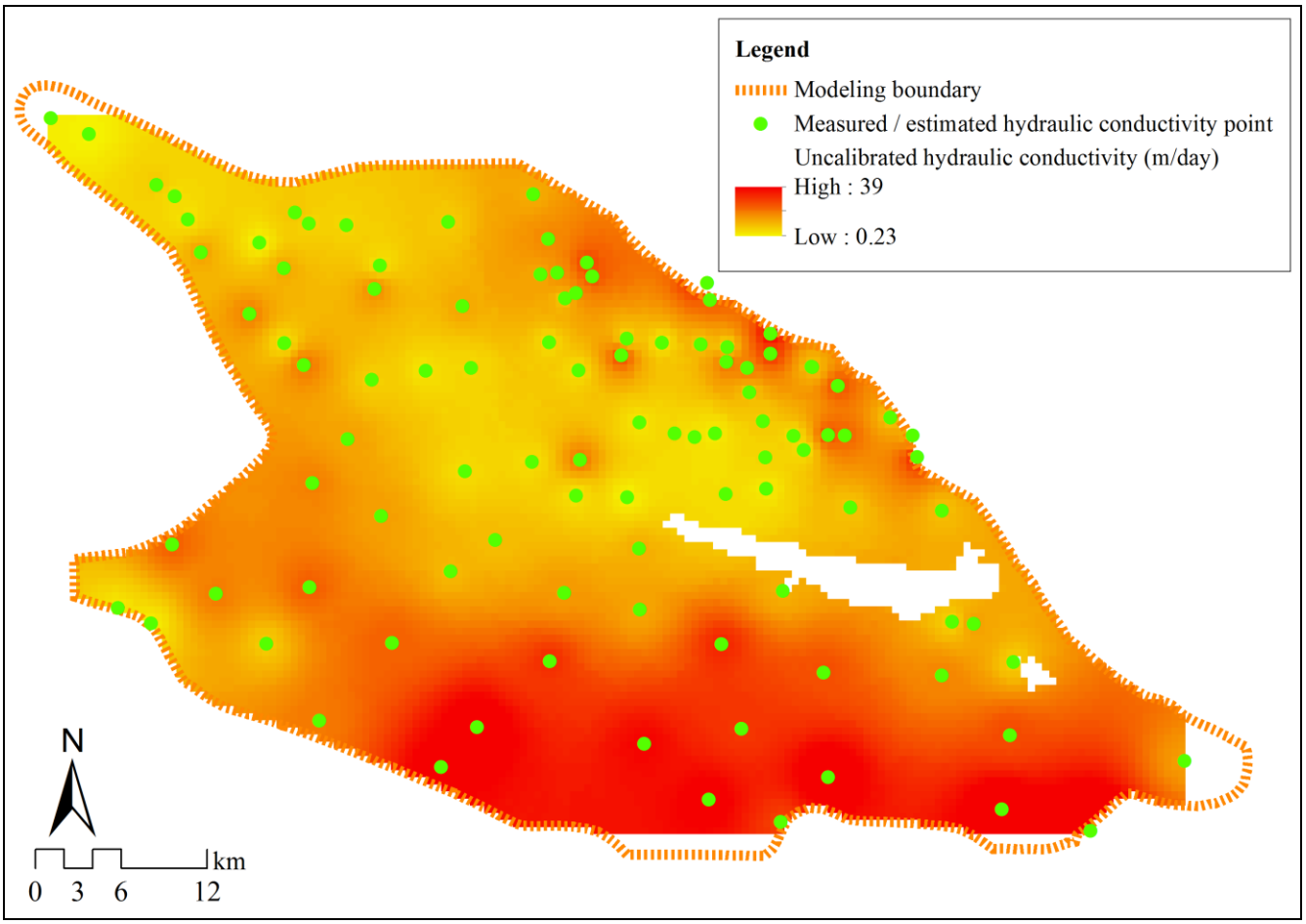


Fig. S5 Uncalibrated (initial) hydraulic conductivity within the aquifer and spatial distribution of measured and estimation points

Specific yield also has the same trend as hydraulic conductivity. The north and south of the plain have the greatest specific yield values (0.169); while, the east and west of plain are assigned the smallest values (0.05) (Fig. S6).

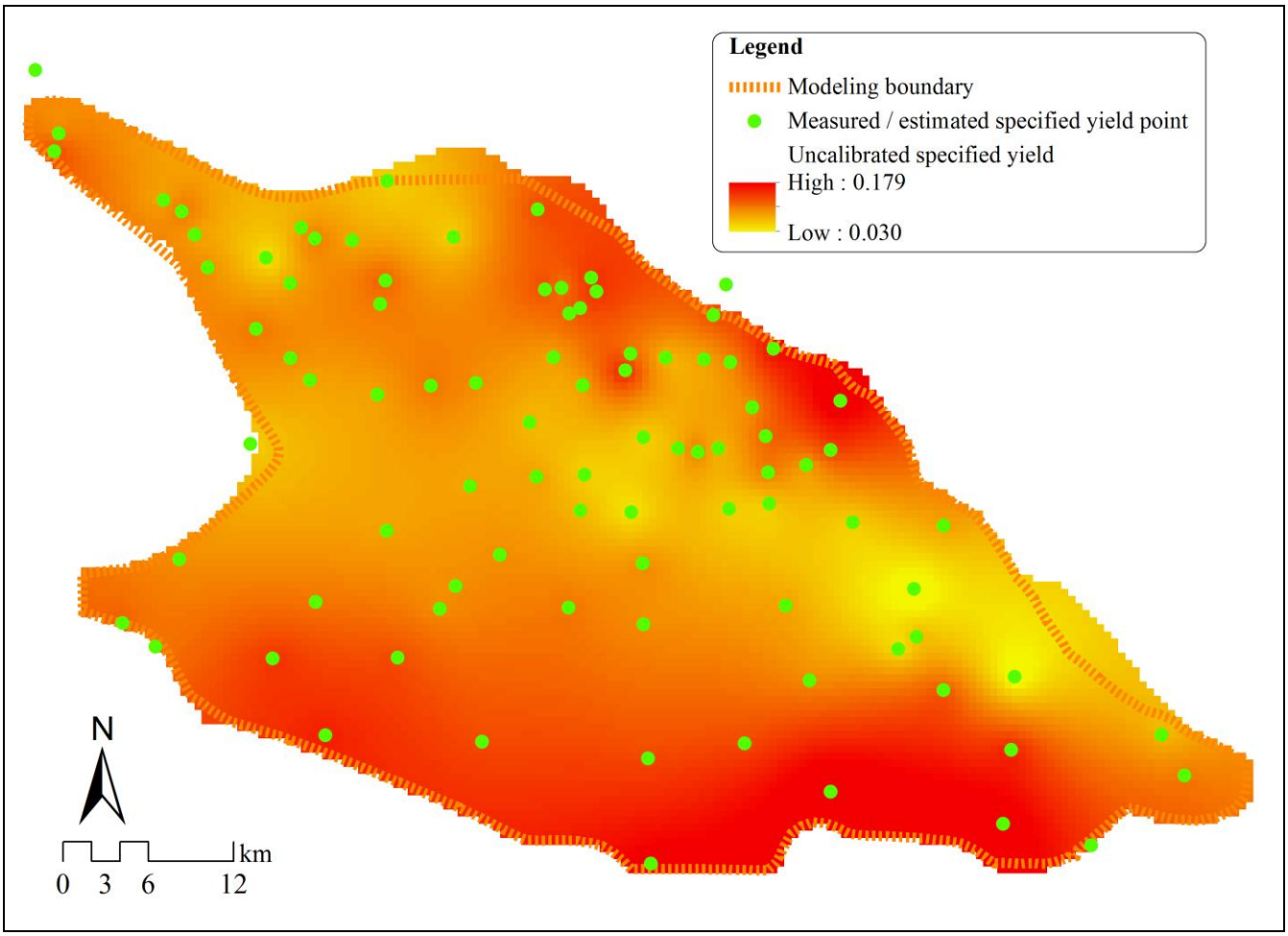


Fig. S6 Uncalibrated (initial) specific yield within the aquifer and spatial distribution of measured and estimation points

2. Surface-water conceptual model

2.1. Climate data

The raw data were inspected carefully before using them in SWAT. It was found that 45%, 23%, 20% and 12% of precipitation was associated with winter, spring, autumn and summer seasons, respectively. The relationship of elevation, longitude, latitude, distance from Binalood Mountain, average height, slope and other topographic and physiographic characteristics were investigated with precipitation at a daily, event, monthly, seasonal and annual scale using single and multiple regression. It was observed that there is no relationship between mentioned variables and precipitation at a daily

and event scale for the study area. On monthly and seasonal scales, some relevance between these parameters and precipitation were found, by means of multiple regressions. However, stronger relations were obtained at the annual scale for both single and multiple regressions.

2.2. Crop management data

Table S2 Management practices for irrigated wheat

Year	Operation type	[a]Date	Description
1	Tillage	September 26	Moldboard Plow
1	Tillage	September 27	Leveler
1	Planting	October 4	-
1	Fertilizer	October 5	Phosphate (18-46-00), 150 kg/ha
1	[b]Manual-Irrigation	October 6	-
2	Fertilizer	March 15	Urea, 50 kg/ha
2	Fertilizer	April 9	Urea, 50 kg/ha
2	Fertilizer	April 30	Urea, 50 kg/ha
2	[c]"Harvest and Killing"	July 13	-

[a] The management practices are based on the average long-term data and these dates were used for all simulation years.

[b] Irrigation interval was considered every 12 days.

[c] The "harvest and kill" operation stops plant growth in the "Hydrologic Response Unit" (HRU).

[d] Moldboard Plow and Leveler are tillage device.

Table S3 Management practices for rainfed wheat

Year	Operation type	Date	Description
1	Tillage	November 23	Moldboard Plow
1	Tillage	November 24	Leveler
1	Planting	December 6	-
1	Fertilizer	December 7	Phosphate (18-46-00), 50 kg/ha
1	Fertilizer	December 7	Urea, 50 kg/ha
2	Harvest & Killing	July 17	-

Table S4 Management practices for corn silage

Year	Operation type	Date	Descriptions
1	Tillage	June 4	Moldboard Plow
3	Planting	June 10	-
4	Fertilizer	June 10	Phosphate (18-46-00), 200 kg/ha
5	[a]Manual-Irrigation	June 11	-
6	Fertilizer	July 11	Urea, 150 kg/ha
7	Fertilizer	August 11	Urea, 150 kg/ha
9	Harvest & Killing	September 21	-

[a] Irrigation interval was considered every 10 days.

3. Surface water modeling

3.1. SWAT description

SWAT simulates two aquifers in each subbasin. The shallow aquifer is an unconfined aquifer that contributes to flow in the main channel or reach of the subbasin. The deep aquifer is a confined aquifer. The recharge to both aquifers on a given day is calculated:

$$\omega_{\text{rchrg},i} = (1 - \exp[-1/\delta_{\text{gw}}]).\omega_{\text{seep}} + \exp[-1/\delta_{\text{gw}}].\omega_{\text{rchrg},i-1} \quad \text{(S3)}$$

where $\omega_{\text{rchrg},i}$ is the amount of recharge entering the aquifers on day i (mm), δ_{gw} is the delay time of the overlying geologic formations (days), ω_{seep} is the total amount of water exiting the bottom of the soil profile on day i (mm), and $\omega_{\text{rchrg},i-1}$ is the amount of recharge entering the aquifers on day $i-1$ (mm). The total amount of water exiting the bottom of the soil profile on day i is calculated:

$$\omega_{\text{seep}} = \omega_{\text{perc},ly=n} + \omega_{\text{crk,btm}} \quad \text{(S4)}$$

where ω_{seep} is the total amount of water exiting the bottom of the soil profile on day i (mm), $\omega_{\text{perc},ly=n}$ is the amount of water percolating out of the lowest layer, n, in the soil profile on day i (mm), and

$\omega_{crk,btm}$ is the amount of water flow past the lower boundary of the soil profile due to bypass flow on day i (mm).

A fraction of the total daily recharge can be routed to the deep aquifer. The amount of water that will be diverted from the shallow aquifer due to percolation to the deep aquifer on a given day is:

$$\omega_{deep} = \beta_{deep}.\omega_{rchrg} \quad \text{(S5)}$$

where ω_{deep} is the amount of water moving into the deep aquifer on day i (mm), β_{deep} is the aquifer percolation coefficient, and ω_{rchrg} is the amount of recharge entering both aquifers on day i (mm). The amount of recharge to the shallow aquifer is:

$$\omega_{rchrg,sh} = \omega_{rchrg} - \omega_{deep} \quad \text{(S6)}$$

where $\omega_{rchrg,sh}$ is the amount of recharge entering the shallow aquifer on day i (mm).

3.2. Model calibration methods

SWAT calibration and validation was done for the period Oct. 2000 to Sep. 2010 and Oct. 2010 to Sep. 2012, respectively. SWAT is usually calibrated based on streamflow data. In this study, data from five gauging stations were used; however spatial distribution of these stations were not suitable for such a vast watershed. Multi-criteria calibration, as suggested by Abbaspour et al. (2007), were adapted to improve the spatial accuracy of the model. Hence, rainfed/irrigated wheat yield data is used for this purpose. Yet, inasmuch as the crop yield data can only provide yearly information, temporal accuracy of the model needs to be enhanced. To overcome this shortcoming, monthly groundwater extraction data were used for calibration of the model, as a novel technique in this study. Different weights for the observed data were considered due to their uncertainty (Abbaspour et al. 2007). Weights of 1, 0.9 and

0.8 were assigned to the monthly measured streamflow, annual estimated crop yield, and monthly estimated groundwater extraction data, respectively.

SWAT was first calibrated with hydrologic parameters, and then these parameters were fixed and calibrated for crop yield and groundwater extraction, in an iterative manner. For this reason, hydrometric stations located in mountainous region of the watershed (namely: Andarab, Bar, Eishabad and Kharvm) were calibrated separately. Next, the whole watershed was calibrated by considering fixed hydrologic parameters for the mentioned stations. Thereafter, subbasins with irrigated and rainfed landuses were calibrated separately based on crop yield and groundwater extraction parameters. Then, the entire watershed was calibrated using fixed crop and groundwater extraction parameters. Lastly, calibration was rechecked by taking into consideration the fixed hydrologic, crop and groundwater extraction parameters.

Different criteria were used in order to evaluate the effectiveness of the model and its ability to make predictions in the calibration and validation periods. These included P-factor, R-factor, Coefficient of Determination (R^2), Nash-Sutcliffe Coefficient (NS) (Nash and Sutcliffe 1970), and Root Mean Square Error (RMSE). In SUFI-2, parameter uncertainty represents all sources of uncertainties such as uncertainty in the driving variables (e.g. rainfall), conceptual model, parameters, and measured data. The degree to which all uncertainties are recognized is measured by an index referred to as the P-factor, which is the percentage of measured data bracketed by the 95% prediction uncertainty (95PPU) and is also known as “predictive coverage”. The 95PPU is calculated at between 2.5% and 97.5% levels of cumulative distribution of the output variable obtained by Latin hypercube sampling. As all actual processes are reflected in the measurements (e.g., discharge), to the degree that the 95PPU brackets the measured data it accounts for the uncertainties. Hence, the P-factor is used as a measure of the uncertainties captured by the parameters ranges. Another measure to quantify the strength of a

calibration/uncertainty analysis is the R-factor, which is the average thickness of the 95PPU band divided by the standard deviation of measured data. SUFI-2 strategy is to bracket most of the measured data (P-factor approaching the maximum value of 100%) with the smallest possible uncertainty band (R-factor approaching the minimum value of zero) (Abbaspour 2007).

The NS coefficient is defined as:

$$NS = 1 - \frac{\sum_{i=1}^{N}(P_i - O_i)^2}{\sum_{i=1}^{N}(P_i - O_{avg})^2} \quad \text{(S7)}$$

where N is the number of values, and P_i and O_i are simulated and observed values, respectively. NS is a normalized statistic that determines the relative magnitude of the residual variance compared to the measured data variance (Nash and Sutcliffe, 1970). It also ranges between $-\infty$ and 1.0, with NS =1 being the optimal value. Values between 0.0 and 1.0 are generally viewed as acceptable levels of performance, whereas values <0.0 indicate unacceptable performance (Moriasi et al. 2007).

4. Groundwater flow calibration

Plots of observed and computed groundwater levels during calibration (Oct. 2000 to Sep. 2010), validation (Oct. 2010 to Sep. 2012) and prediction (Oct. 2012 to Sep. 2025) periods for all observation wells are shown in Fig. S8.

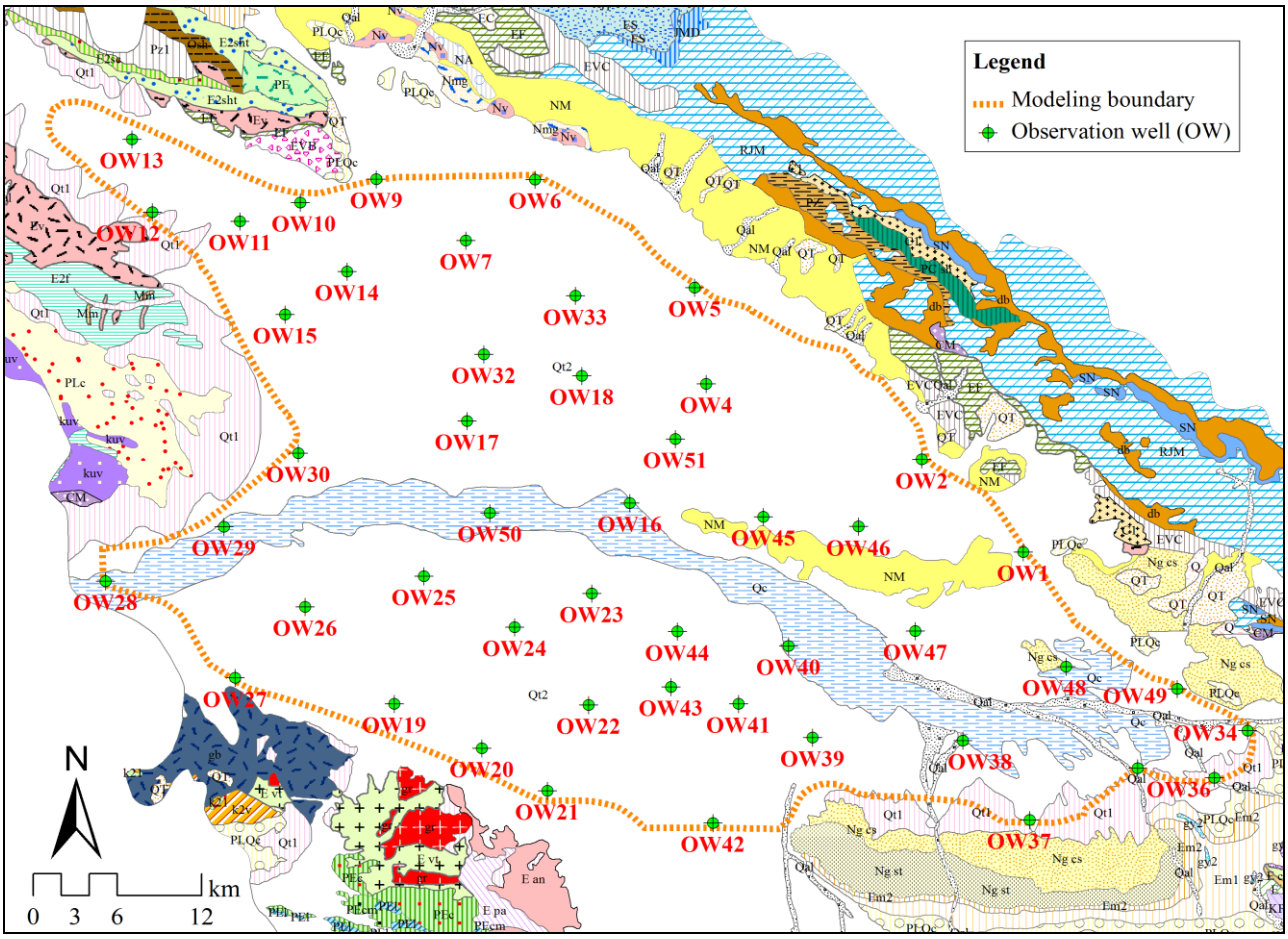
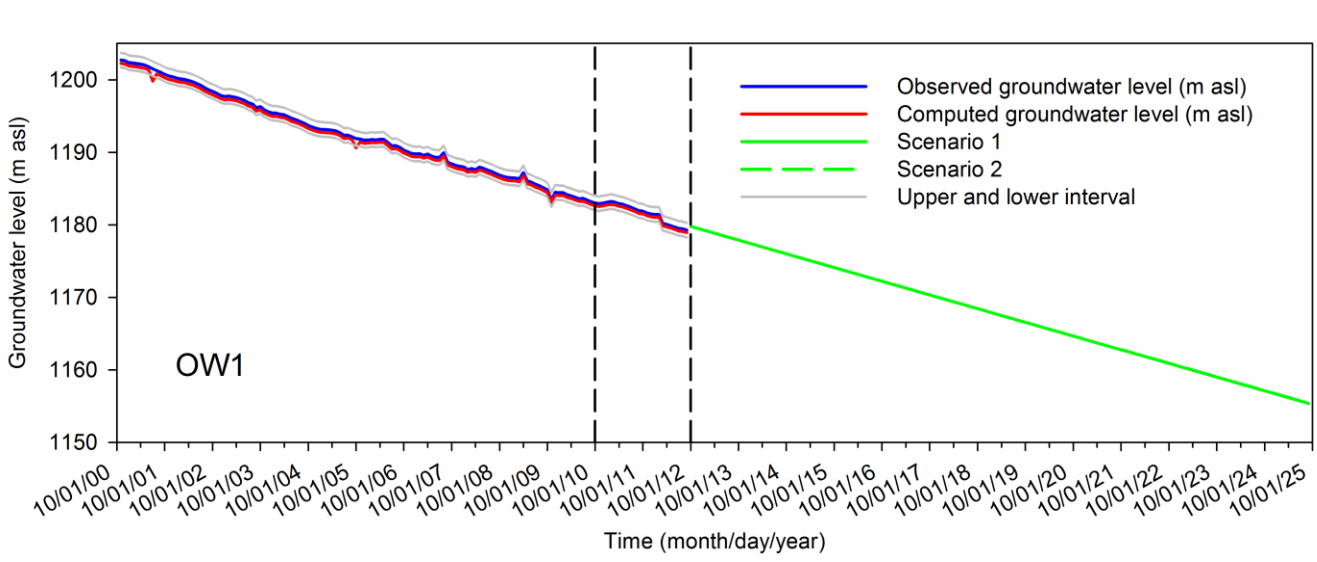
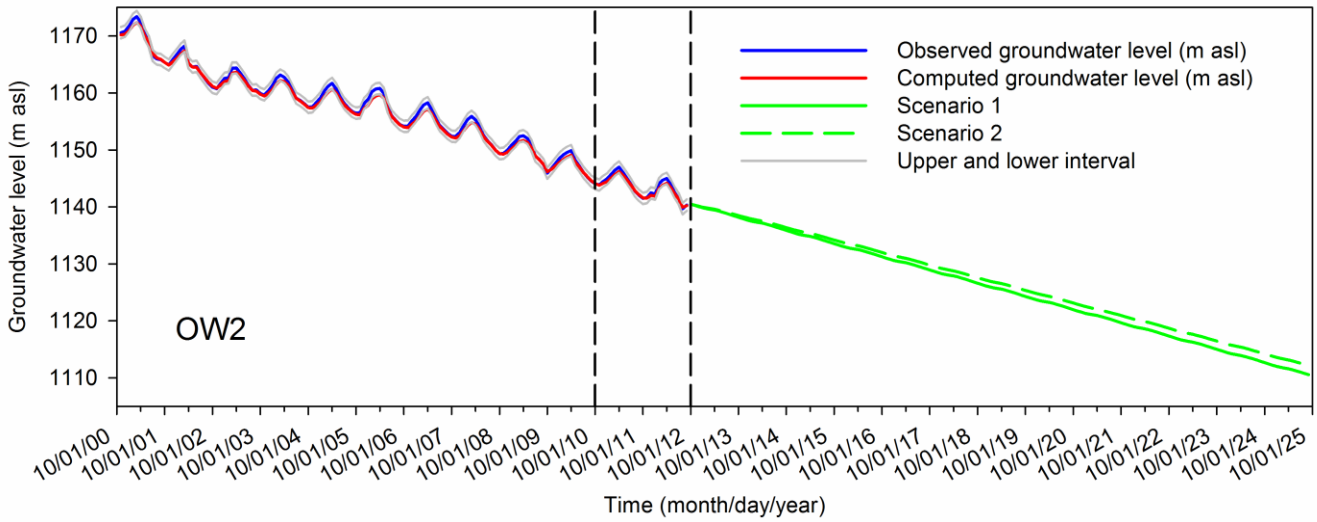
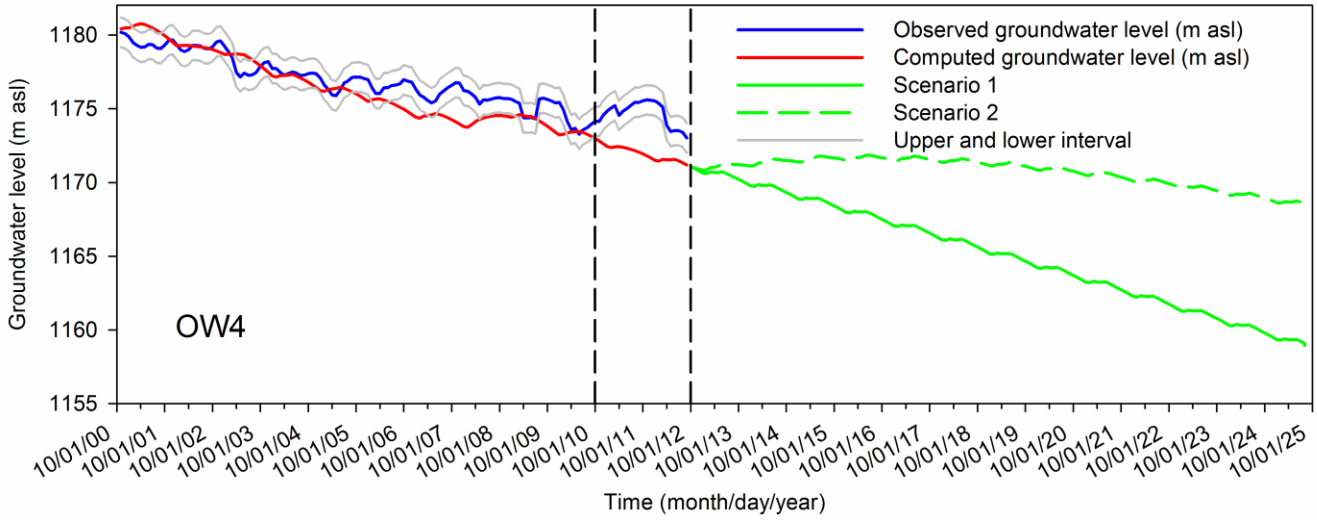
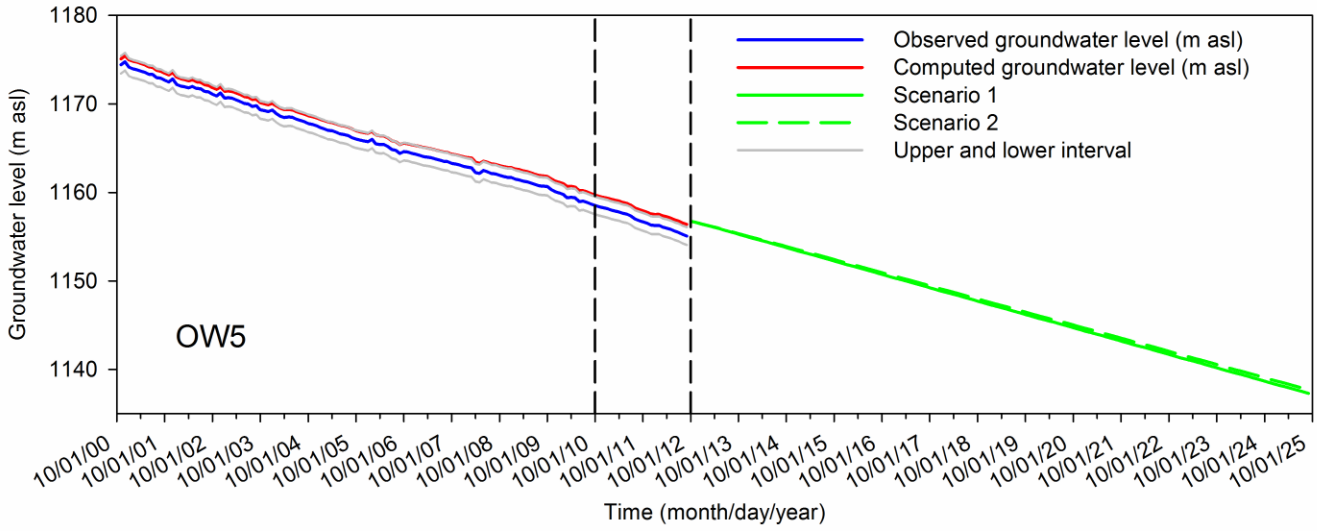


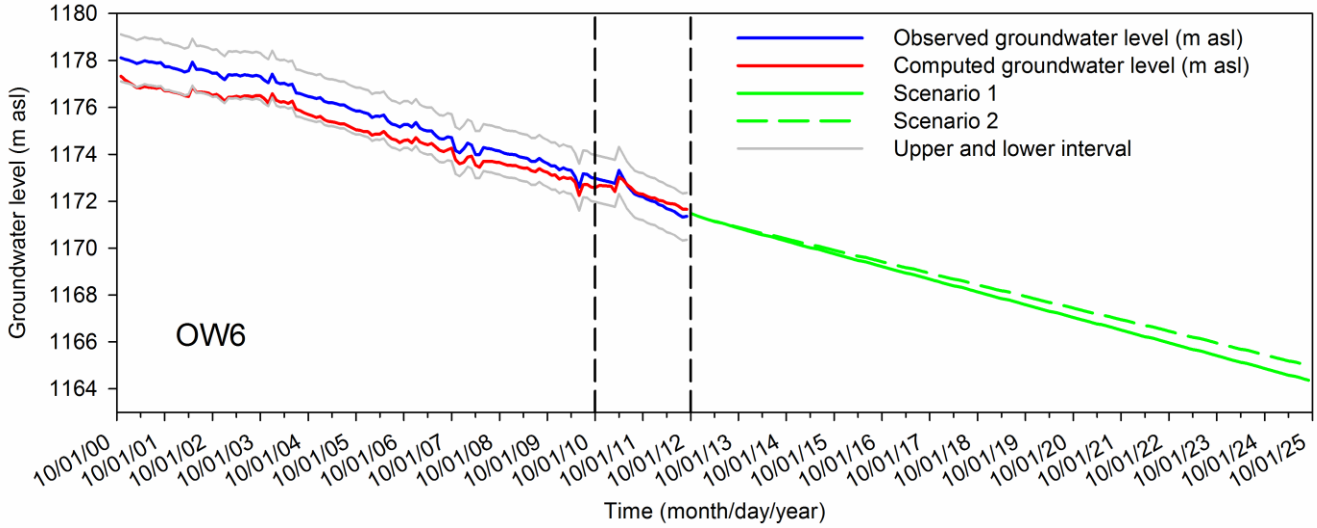
Fig. S7 Location of observation wells within the Neishaboor plain (See main article for description of geology)

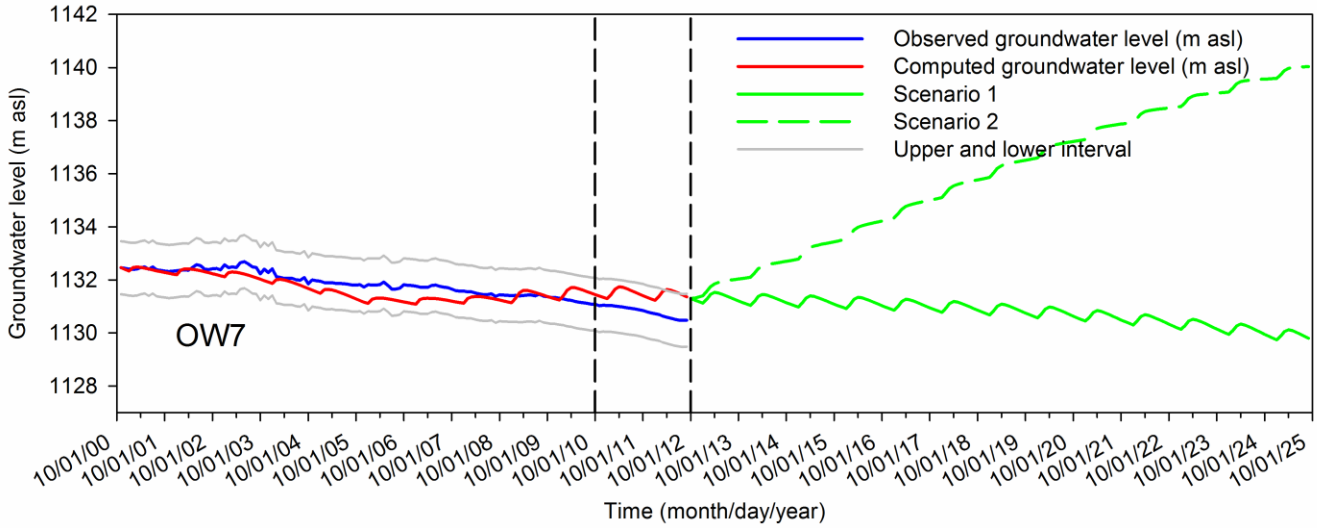


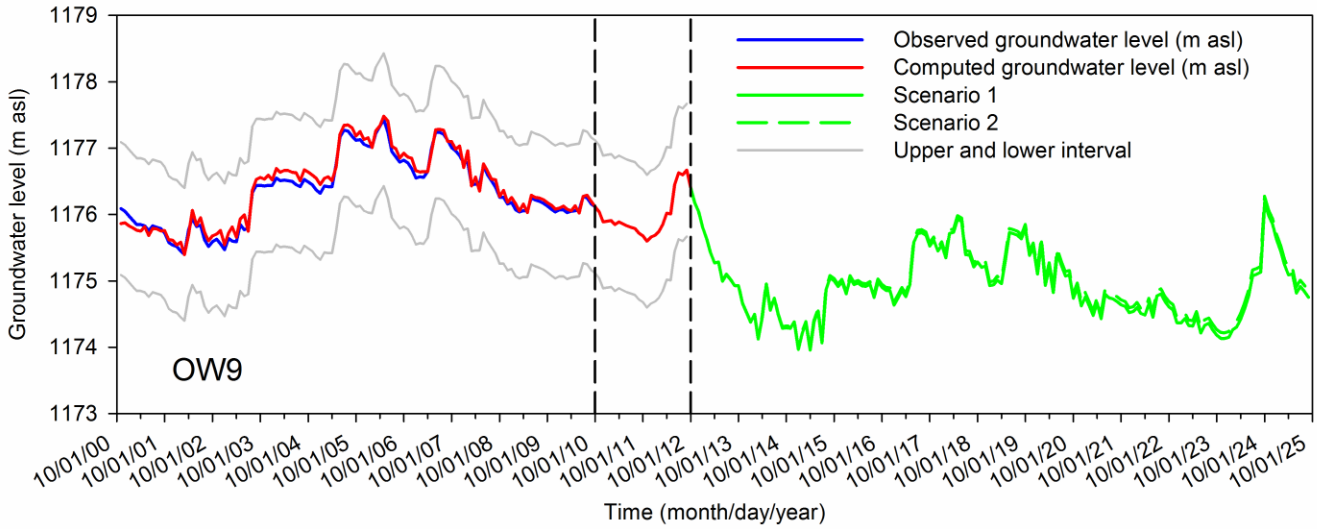

Groundwater level (m asl)
1170
1160
1150
1140
1130
1120
1110
OW2
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
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10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

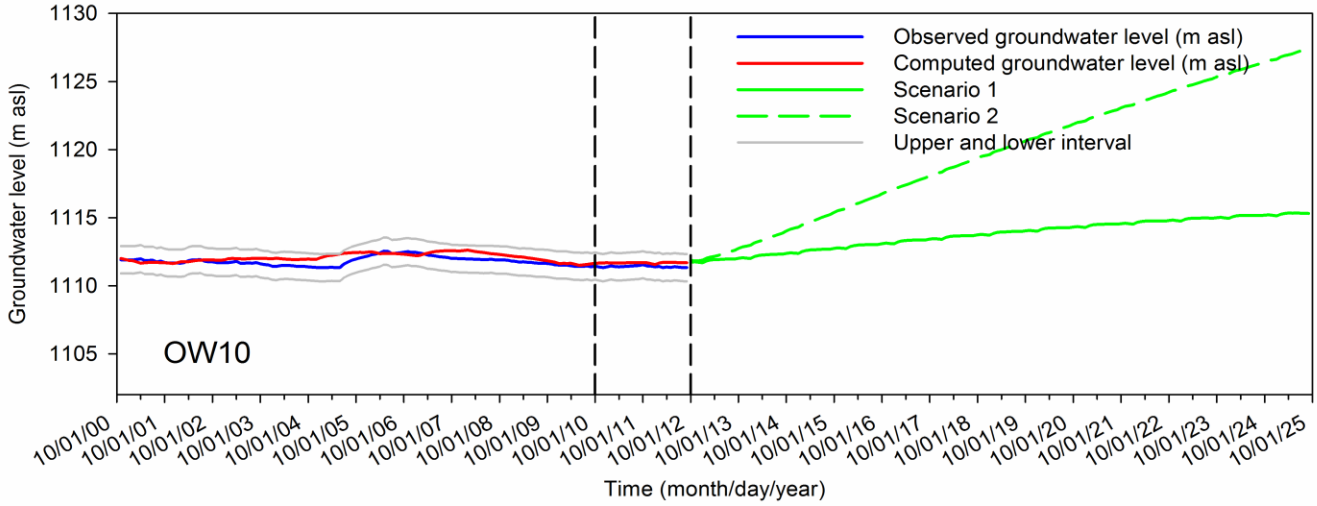

Groundwater level (m asl)
1180
1175
1170
1165
1160
1155
OW4
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
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10/01/24
10/01/25
Time (month/day/year)

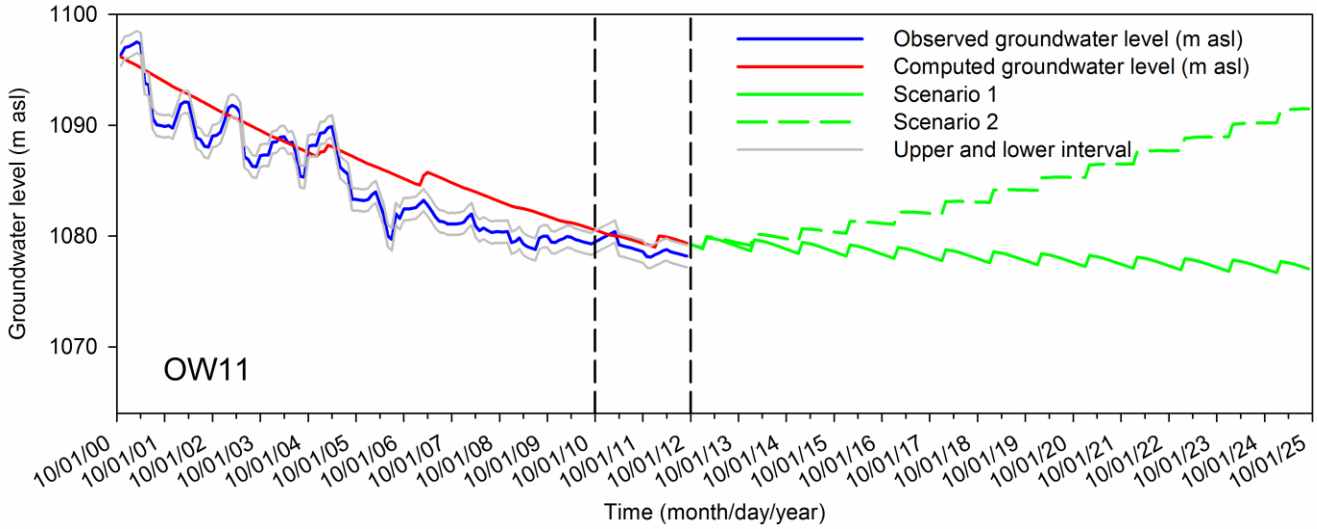

Groundwater level (m asl)
1180
1170
1160
1150
1140
OW5
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
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10/01/25
Time (month/day/year)

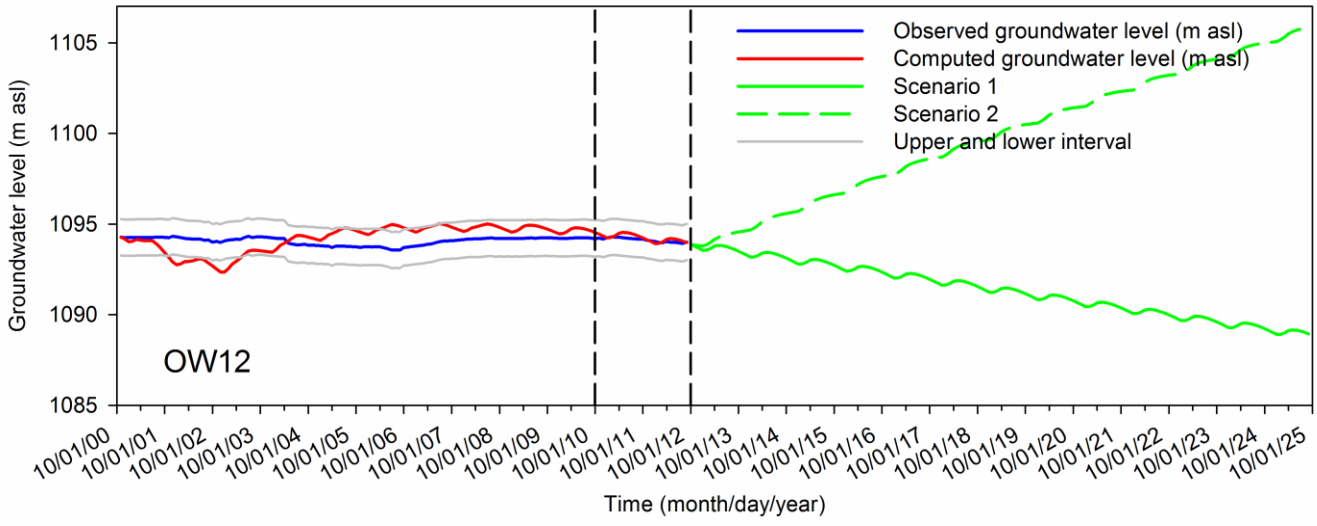

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
OW6
Groundwater level (m asl)
Time (month/day/year)

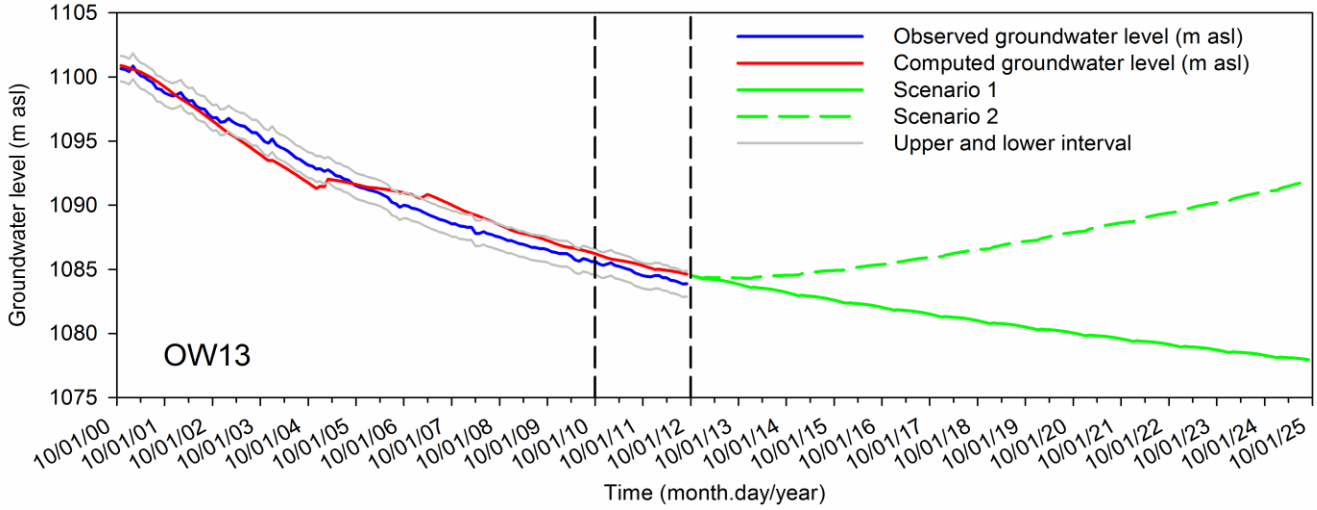

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
OW7
Groundwater level (m asl)
Time (month/day/year)

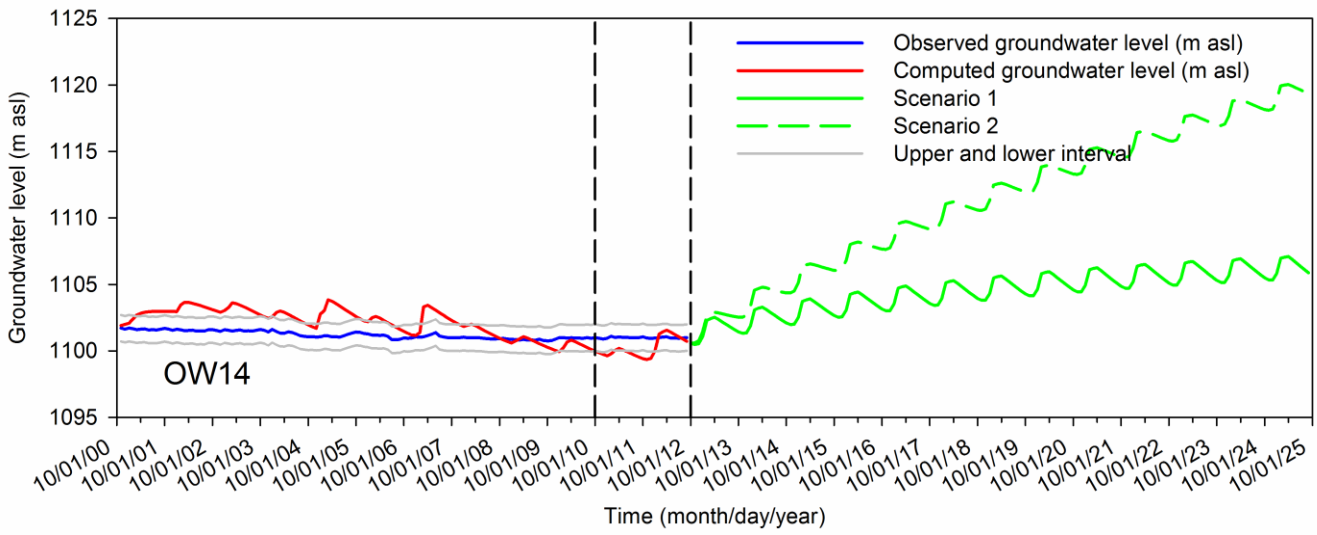

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
OW9
Groundwater level (m asl)
Time (month/day/year)

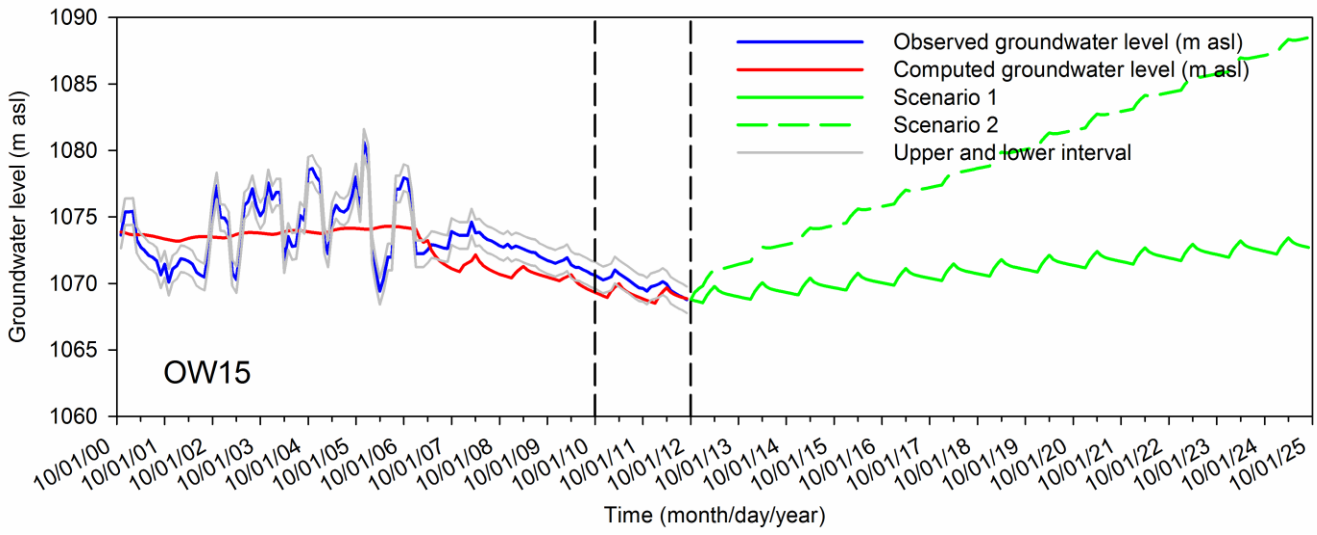

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1130
1125
1120
1115
1110
1105
OW10
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
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10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

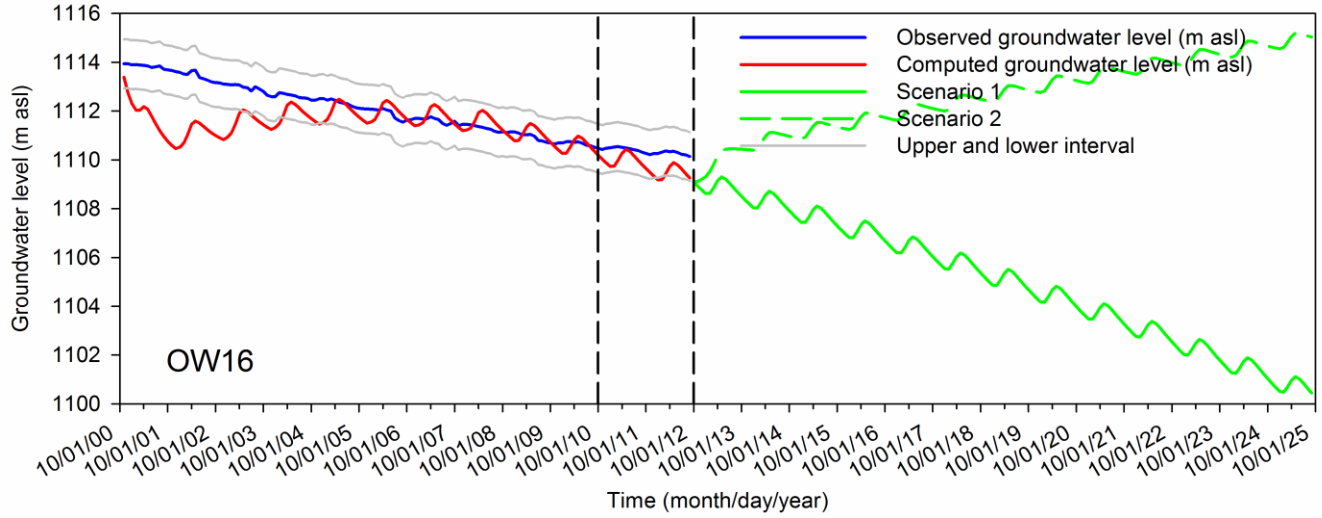

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1100
1090
1080
1070
OW11
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

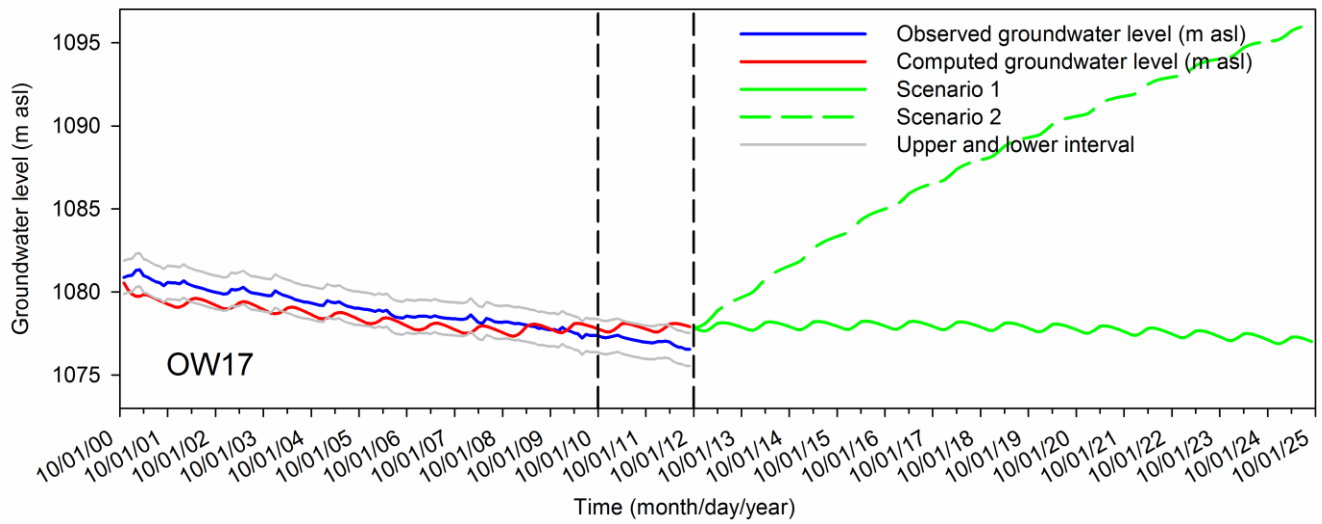

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1105
1100
1095
1090
1085
OW12
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

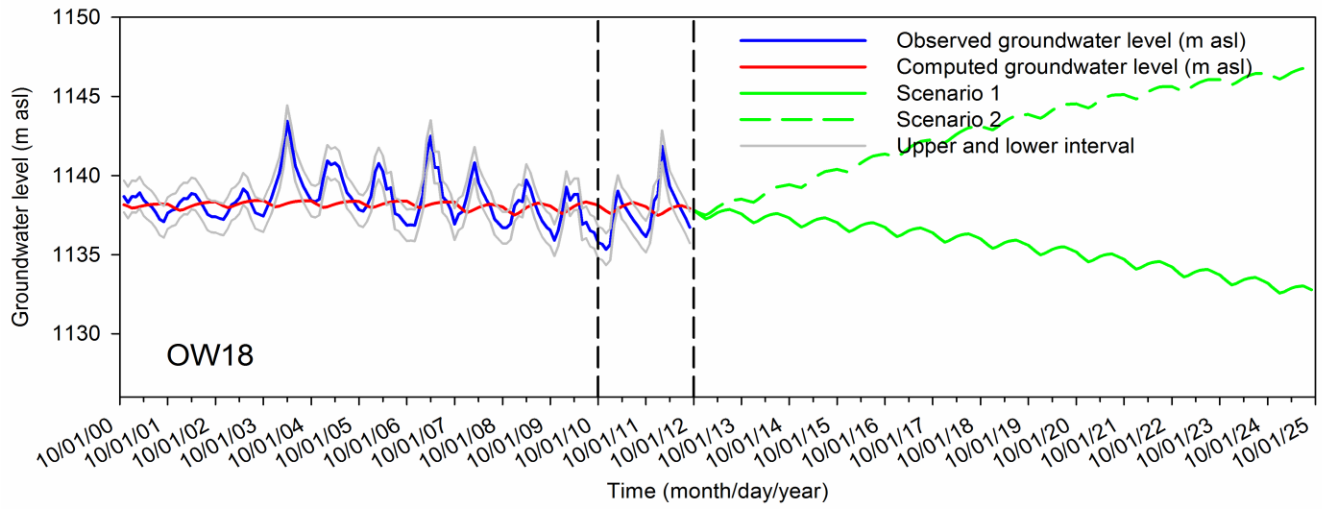

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1105
1100
1095
1090
1085
1080
1075
OW13
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month.day/year)

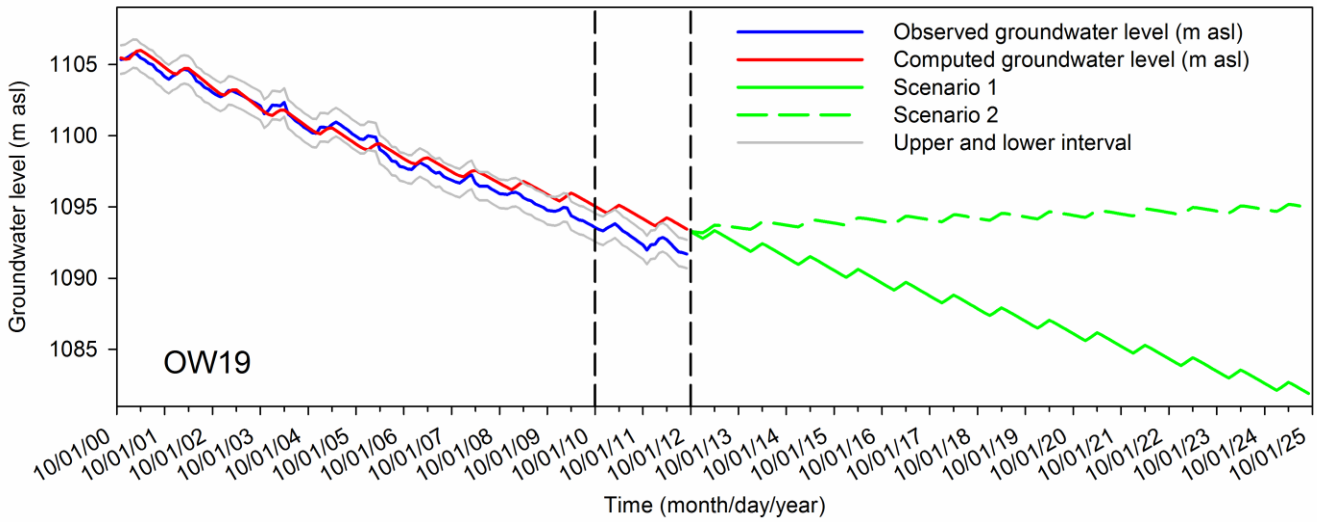

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1125
1120
1115
1110
1105
1100
1095
OW14
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

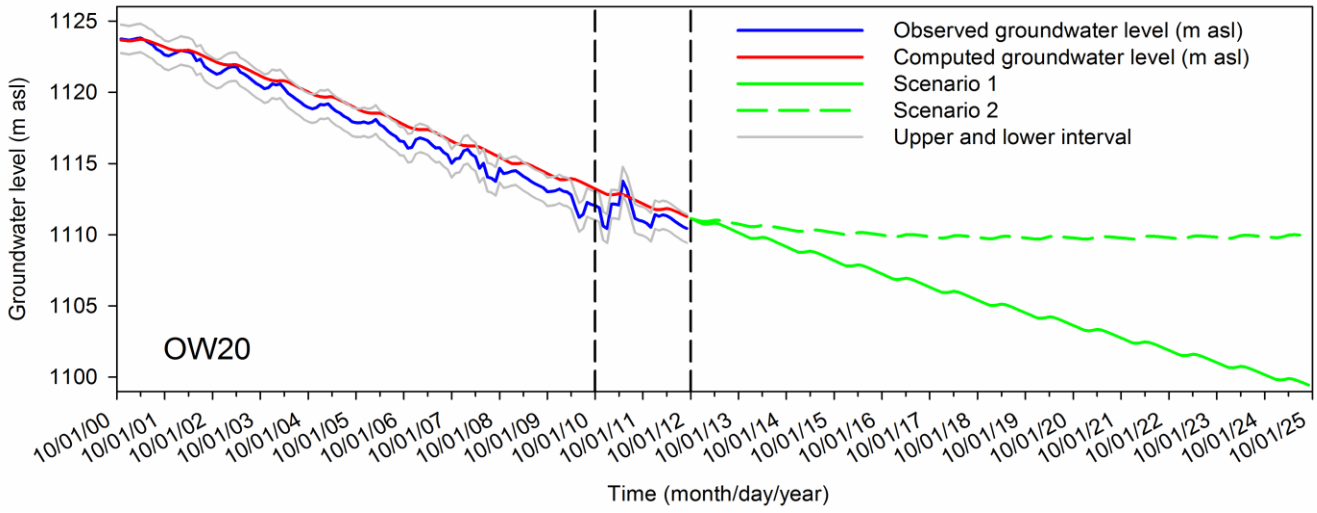

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1090
1085
1080
1075
1070
1065
1060
OW15
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

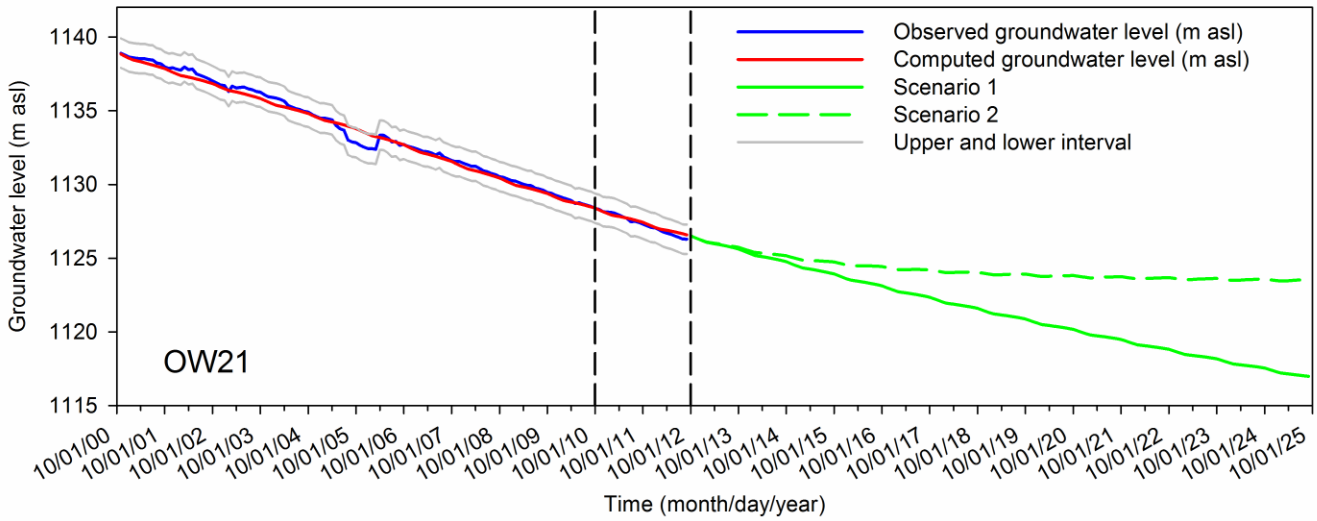

Groundwater level (m asl)
1116
1114
1112
1110
1108
1106
1104
1102
1100
OW16
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

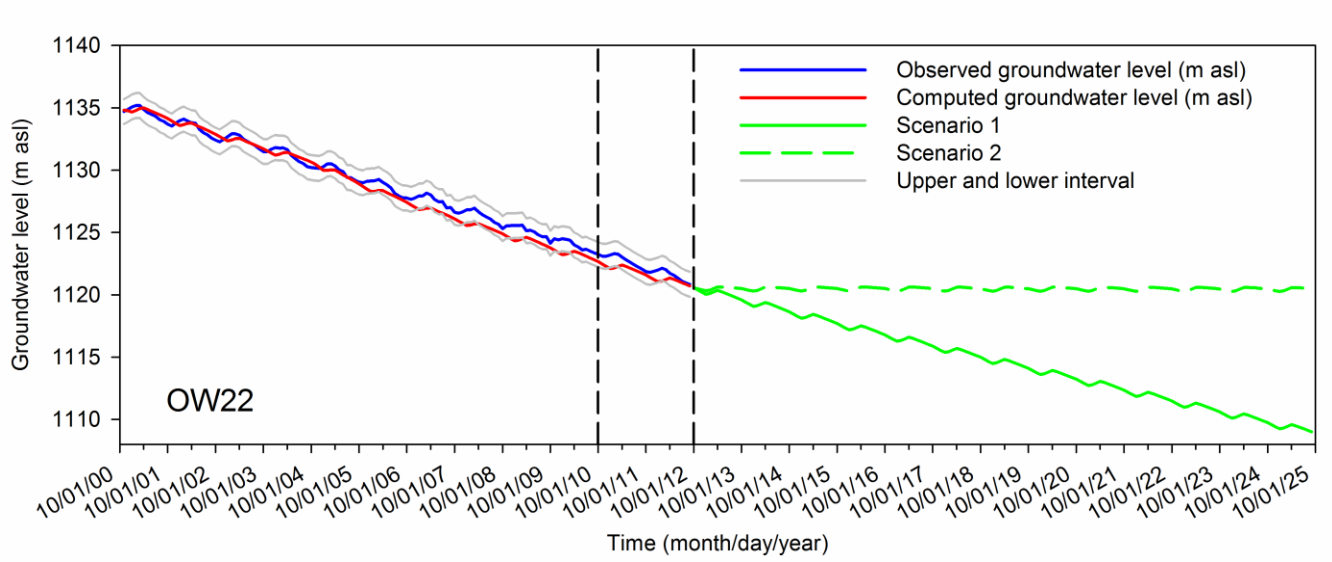

Groundwater level (m asl)
1095
1090
1085
1080
1075
OW17
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

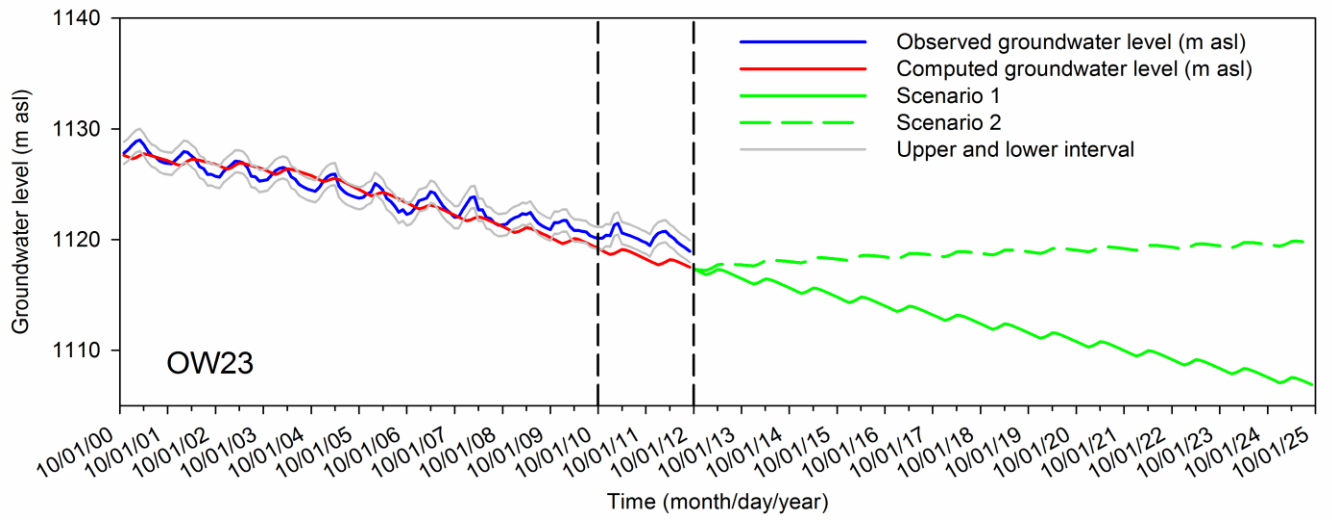

Groundwater level (m asl)
1150
1145
1140
1135
1130
OW18
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

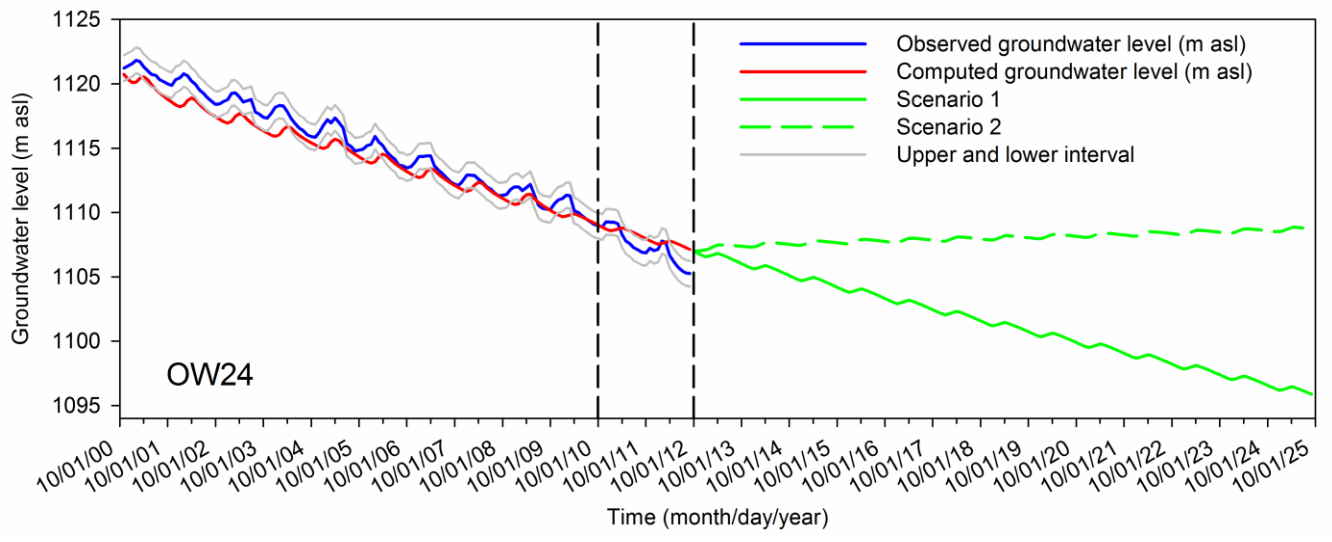

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1105
1100
1095
1090
1085
OW19
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

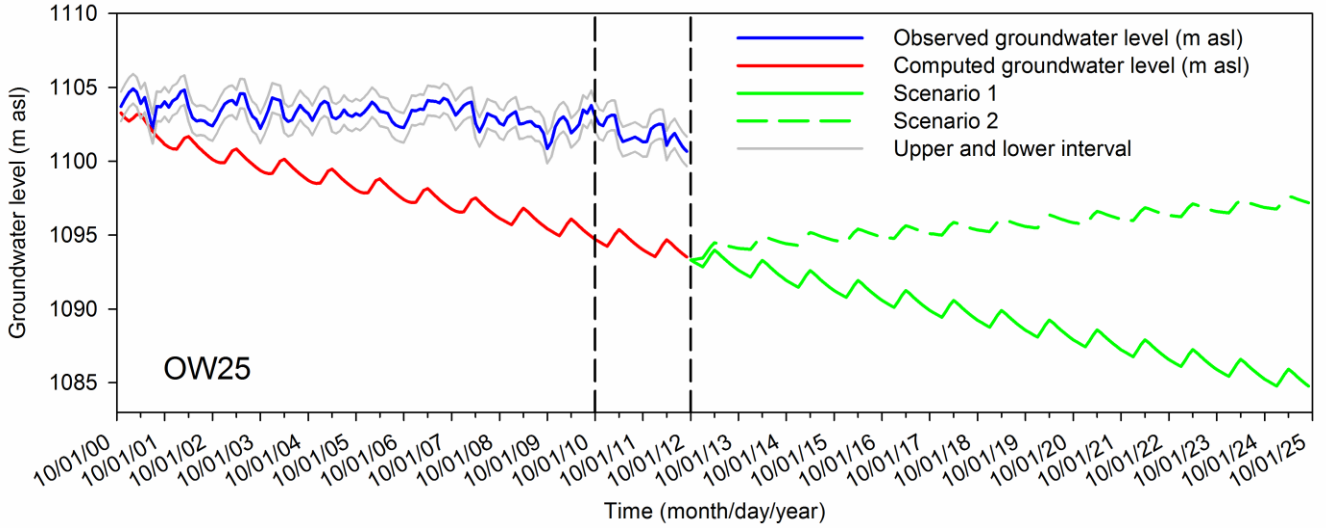

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1125
1120
1115
1110
1105
1100
OW20
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

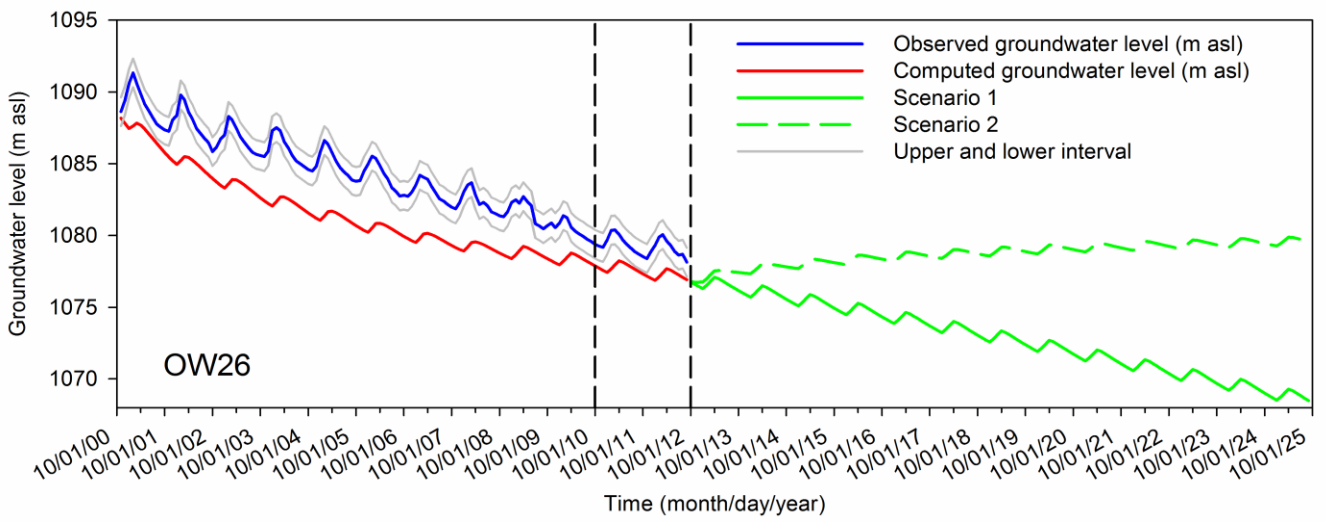

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1140
1135
1130
1125
1120
1115
OW21
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

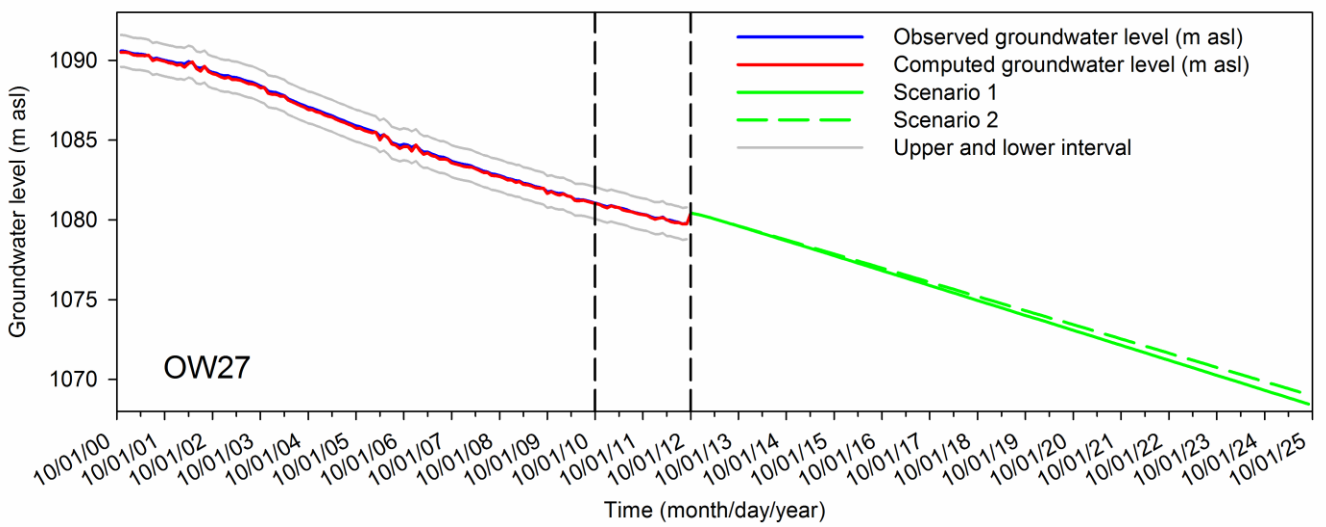

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1140
1135
1130
1125
1120
1115
1110
OW22
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

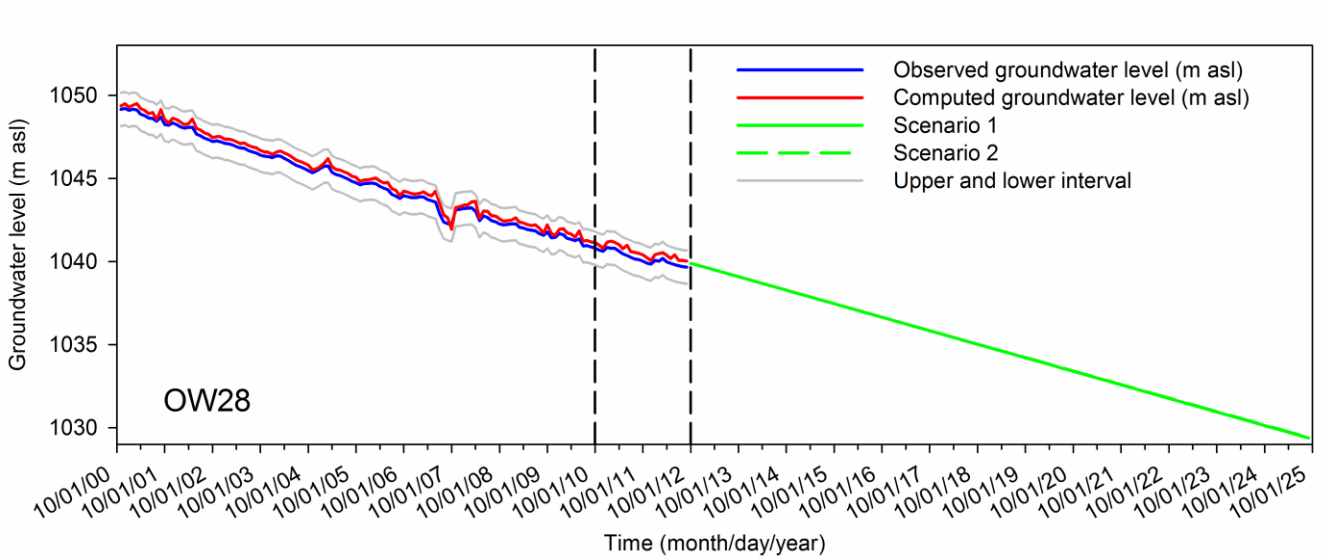

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1140
1130
1120
1110
OW23
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

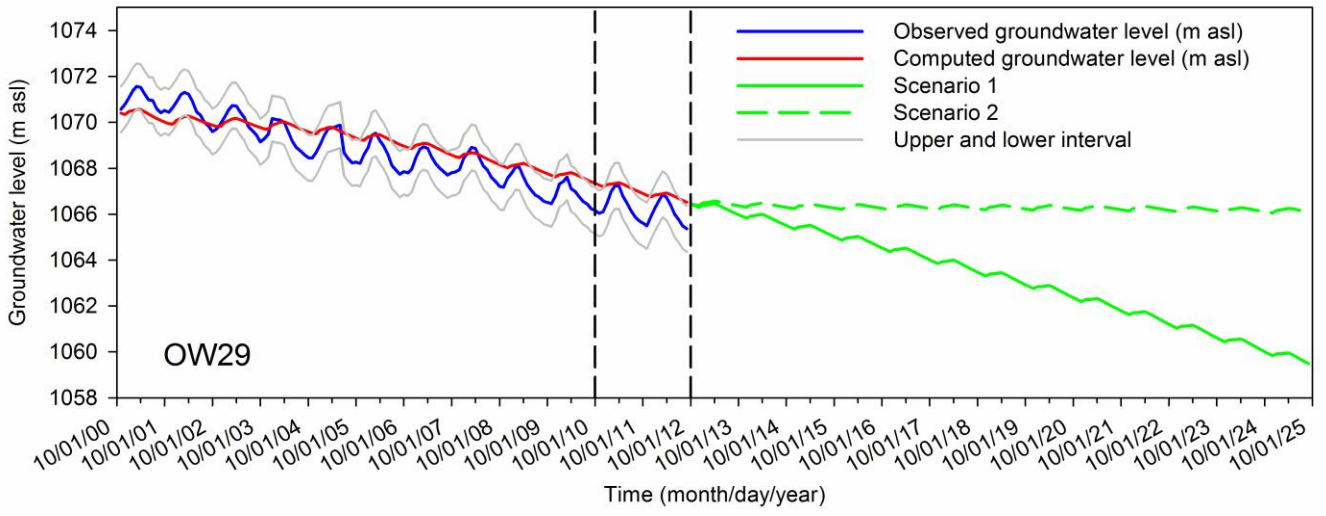

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1125
1120
1115
1110
1105
1100
1095
OW24
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

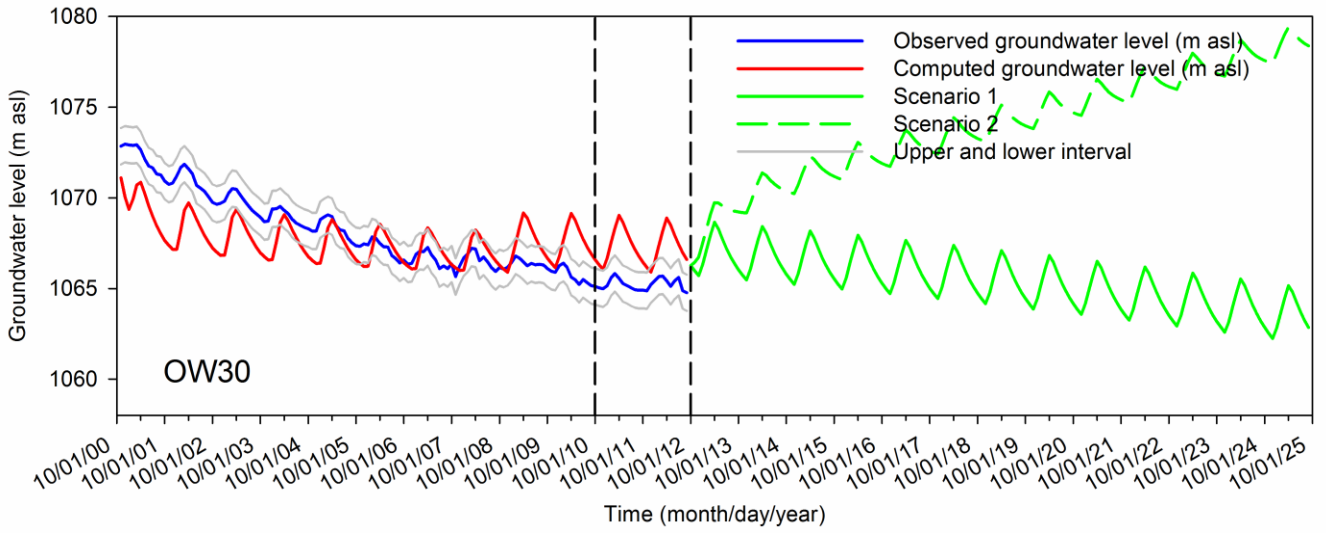

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1110
1105
1100
1095
1090
1085
OW25
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

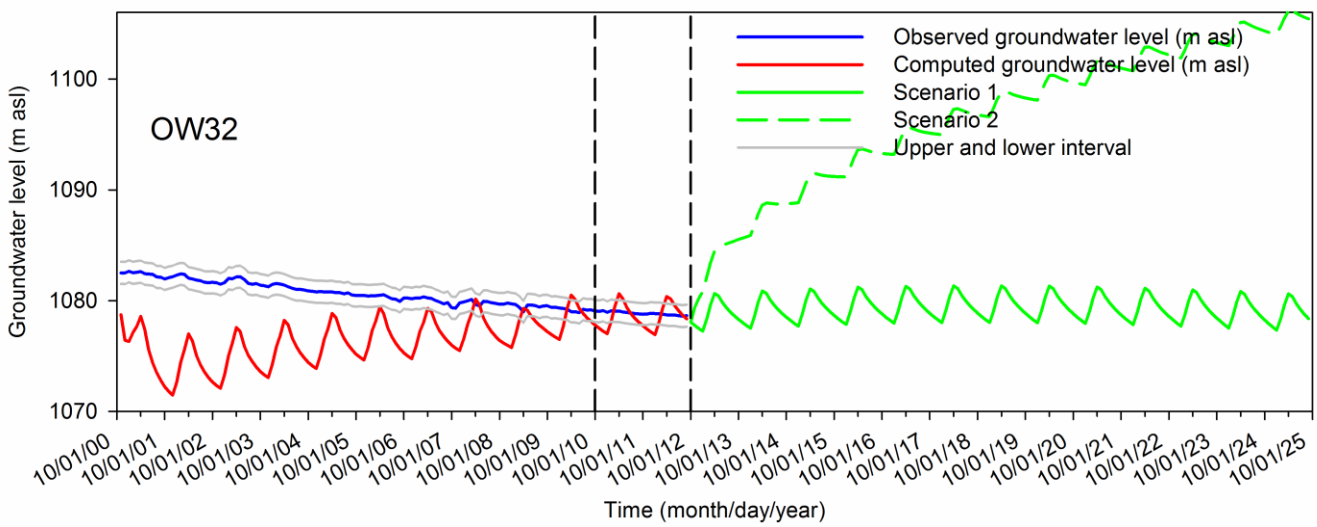

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1095
1090
1085
1080
1075
1070
OW26
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

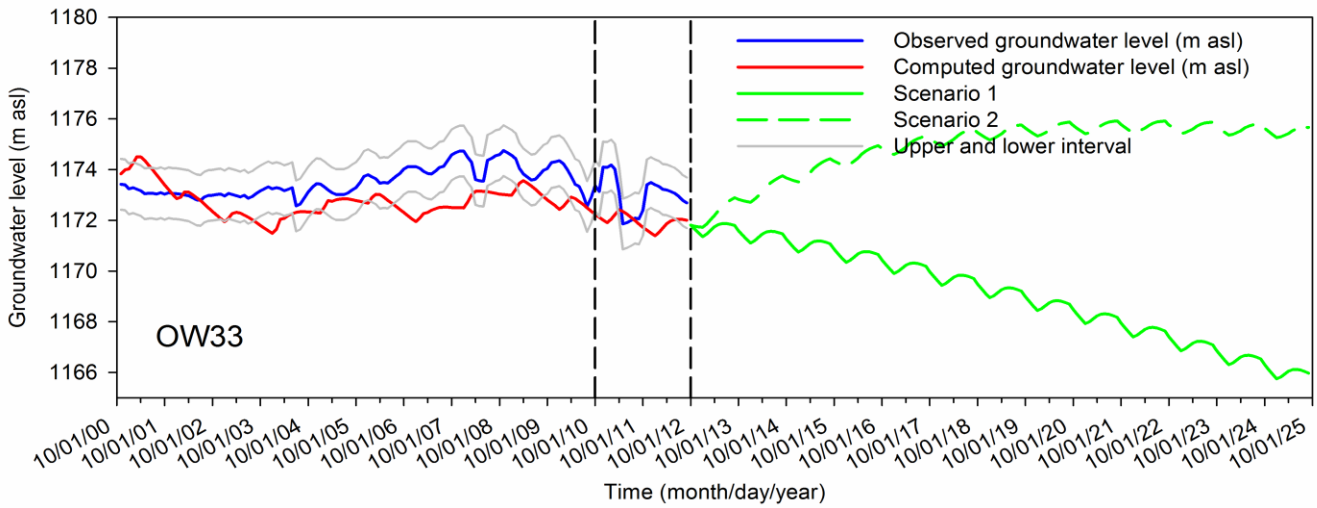

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1090
1085
1080
1075
1070
OW27
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

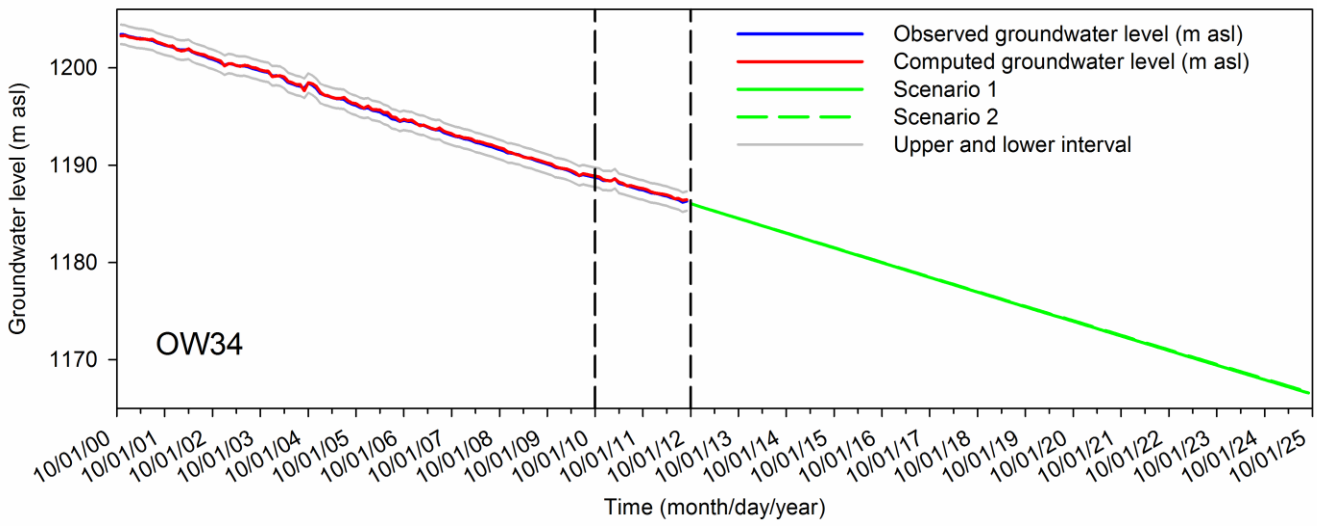

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1050
1045
1040
1035
1030
OW28
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

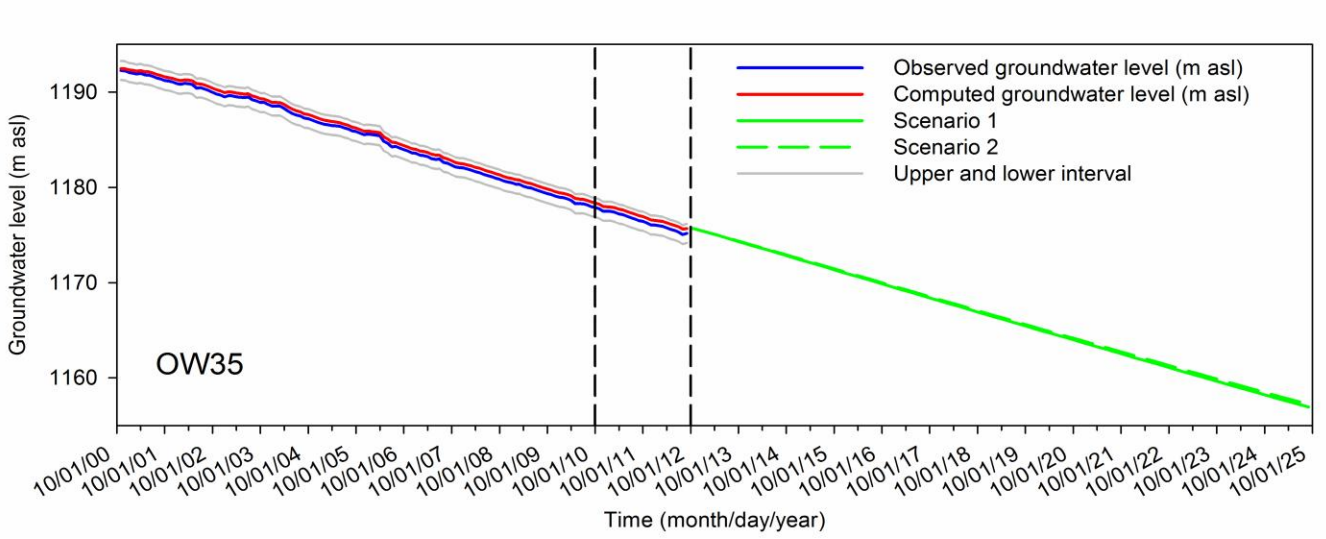

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1074
1072
1070
1068
1066
1064
1062
1060
1058
OW29
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

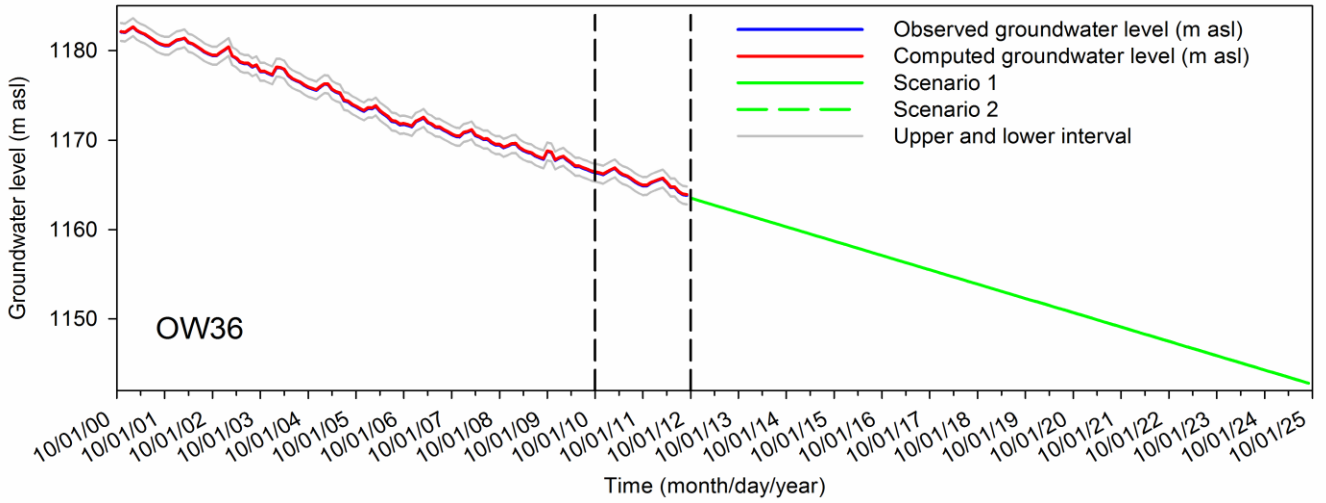

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1080
1075
1070
1065
1060
OW30
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

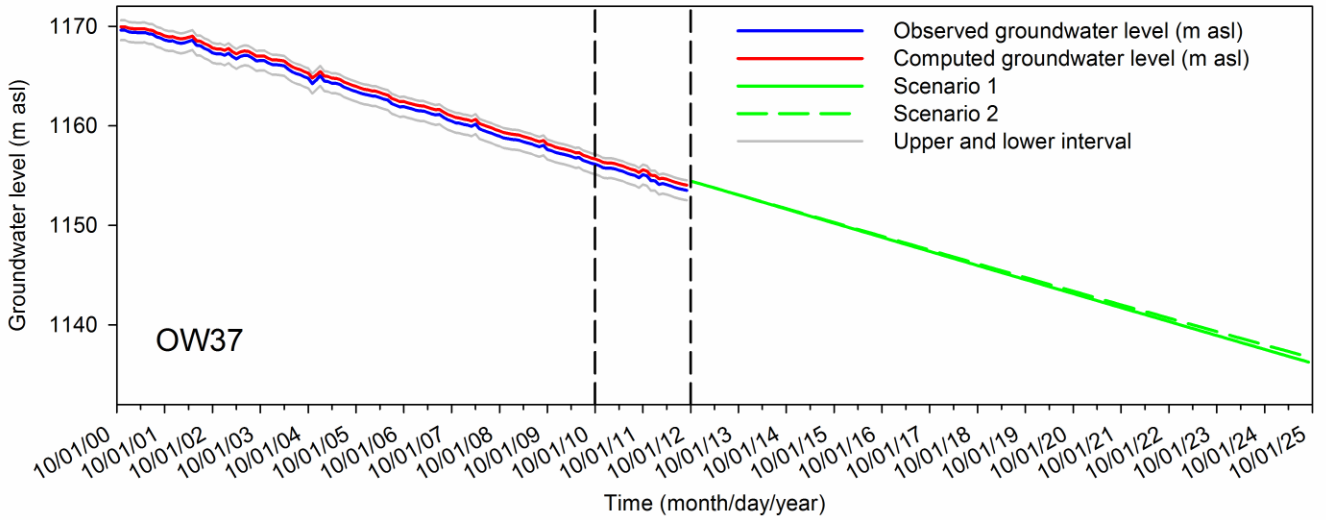

OW32
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1100
1090
1080
1070
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

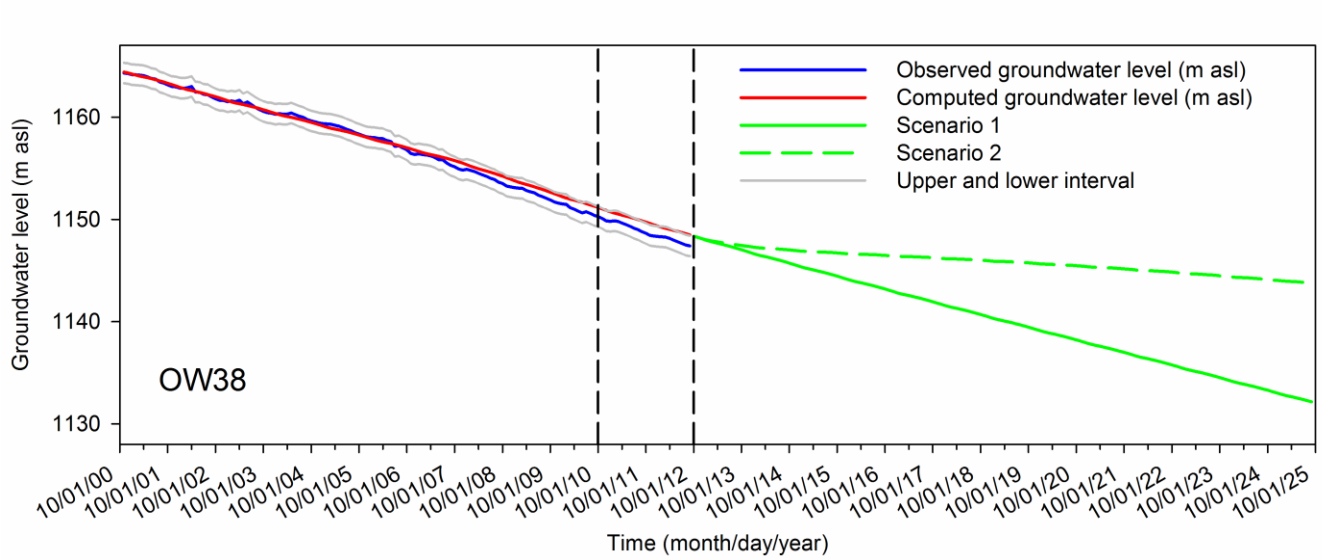

OW33
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1180
1178
1176
1174
1172
1170
1168
1166
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

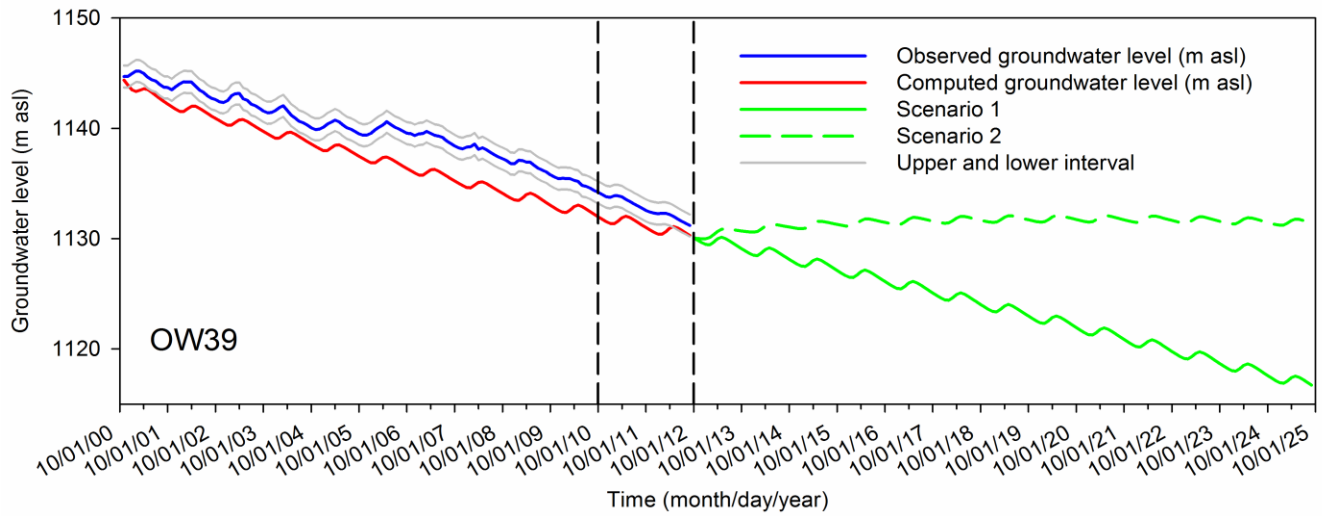

OW34
Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1200
1190
1180
1170
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

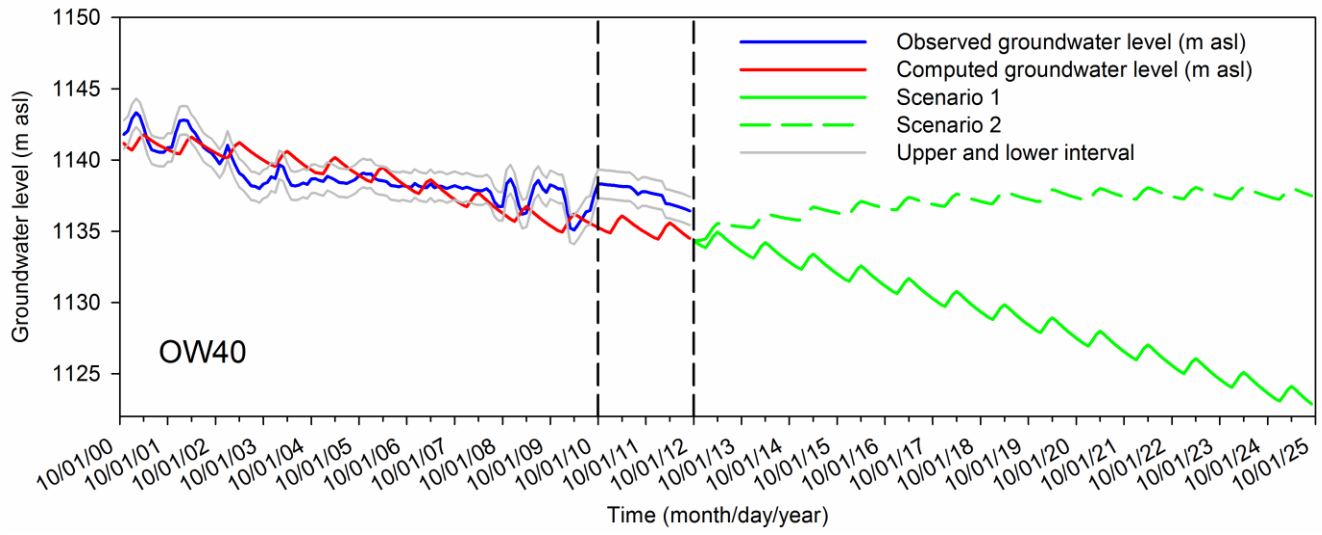

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1190
1180
1170
1160
OW35
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

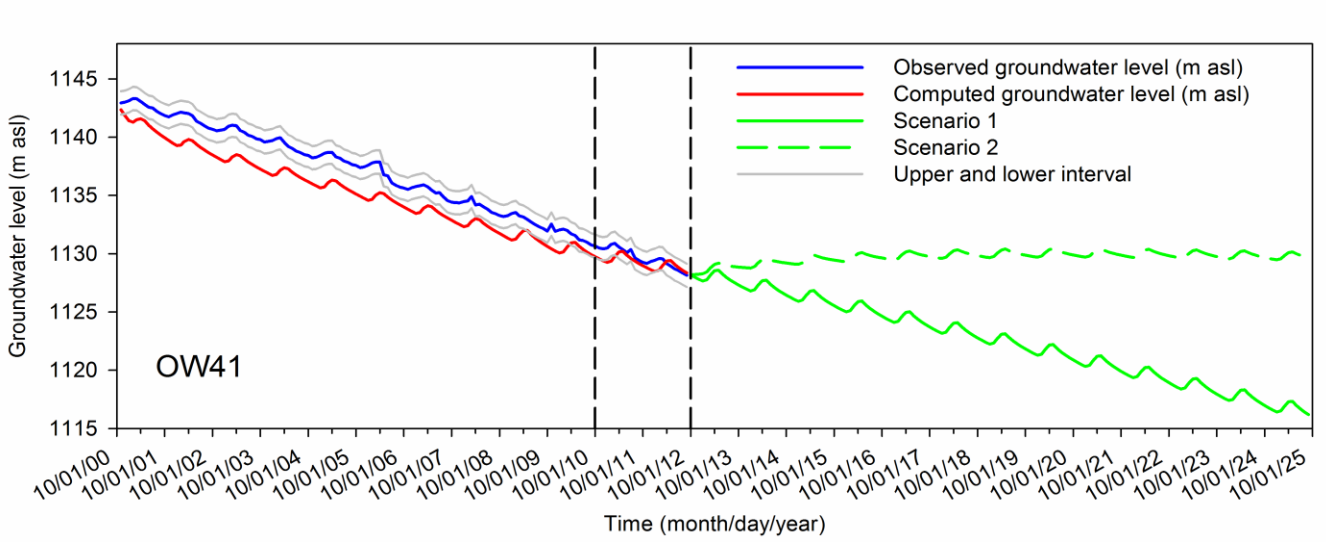

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1180
1170
1160
1150
OW36
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

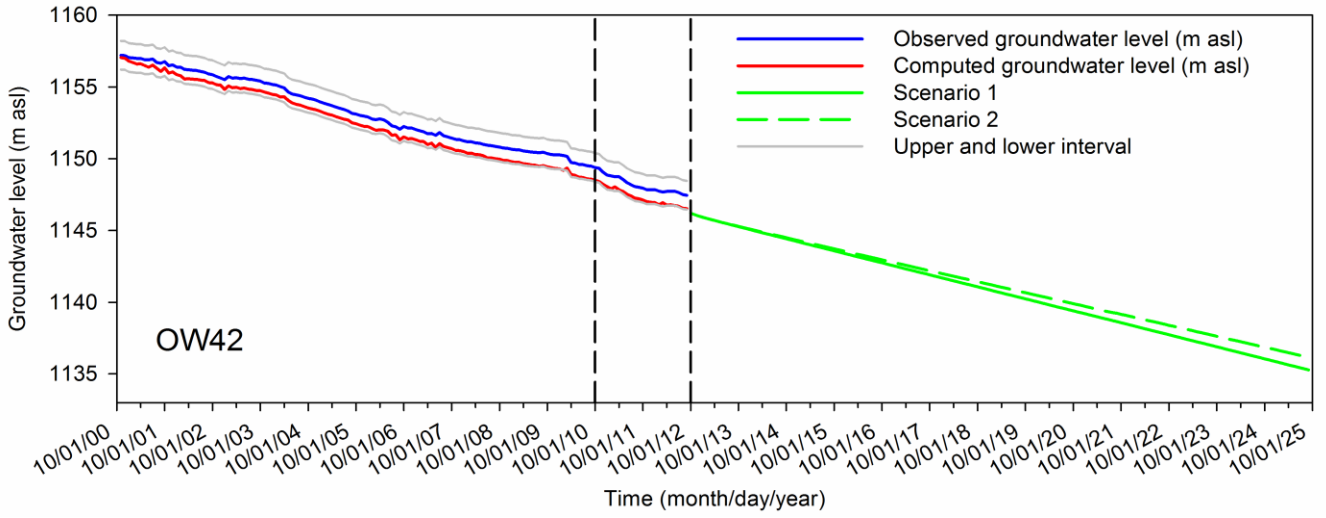

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1170
1160
1150
1140
OW37
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

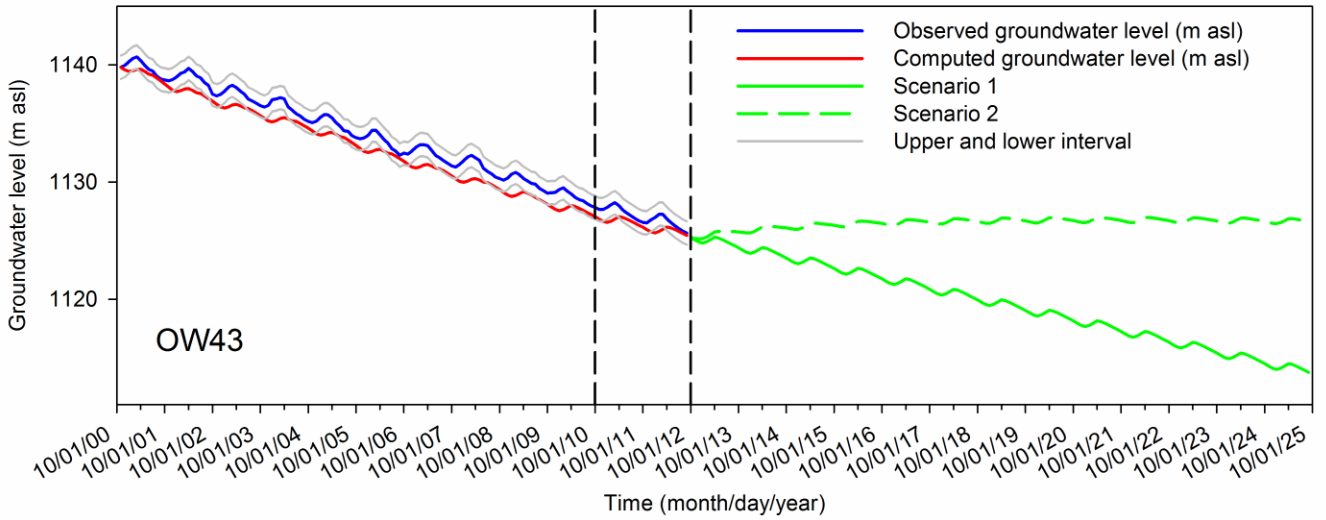

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1160
1150
1140
1130
OW38
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

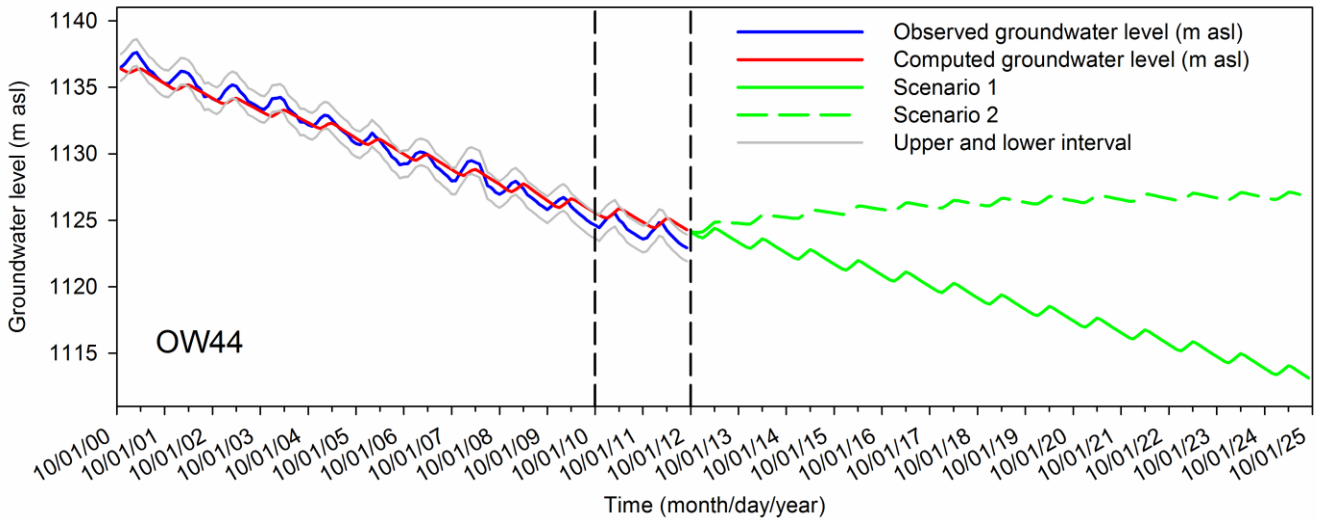

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1150
1140
1130
1120
OW39
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

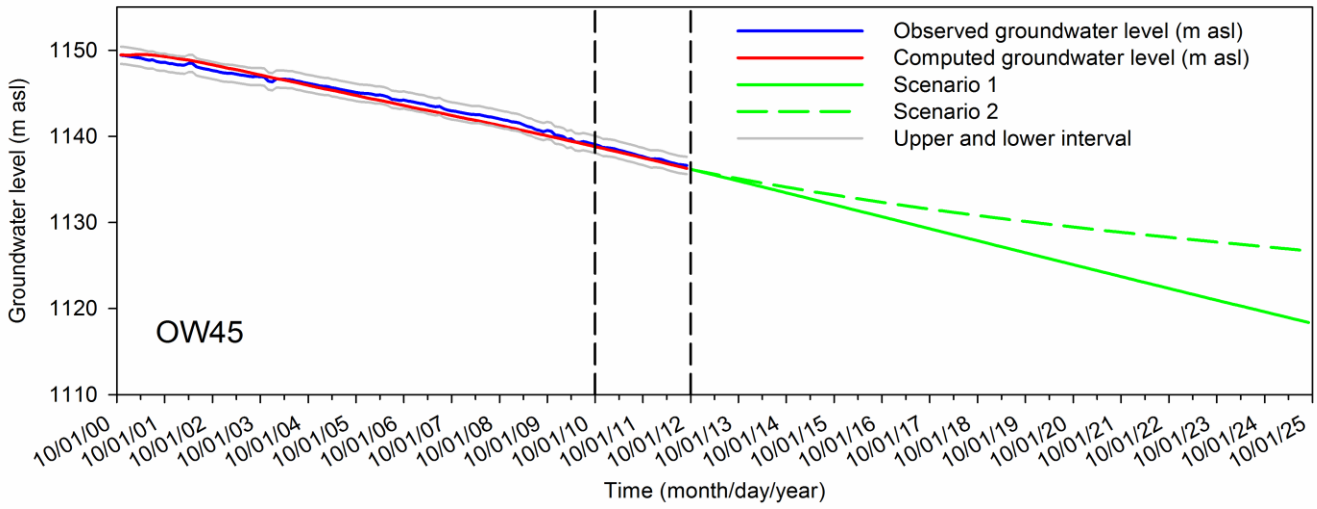

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1150
1145
1140
1135
1130
1125
OW40
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

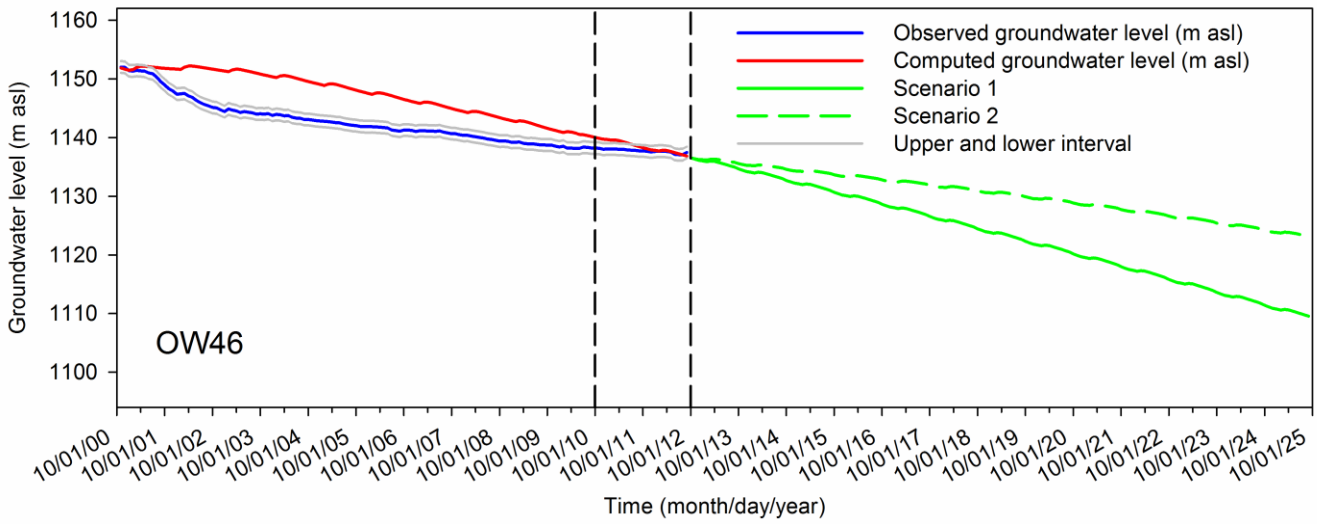

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1145
1140
1135
1130
1125
1120
1115
OW41
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

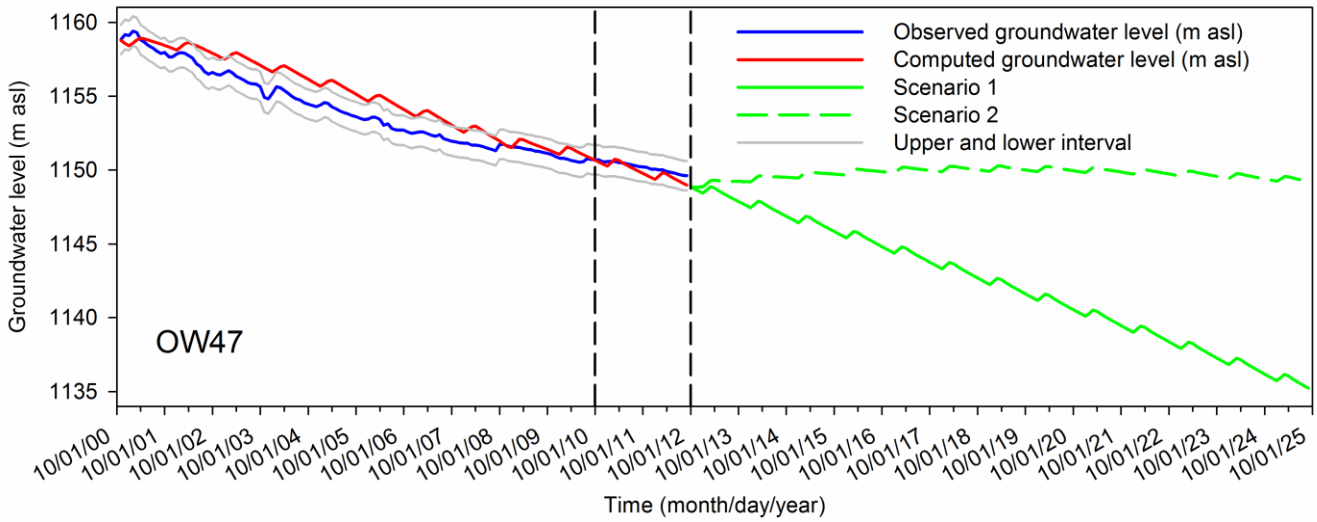

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1160
1155
1150
1145
1140
1135
OW42
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

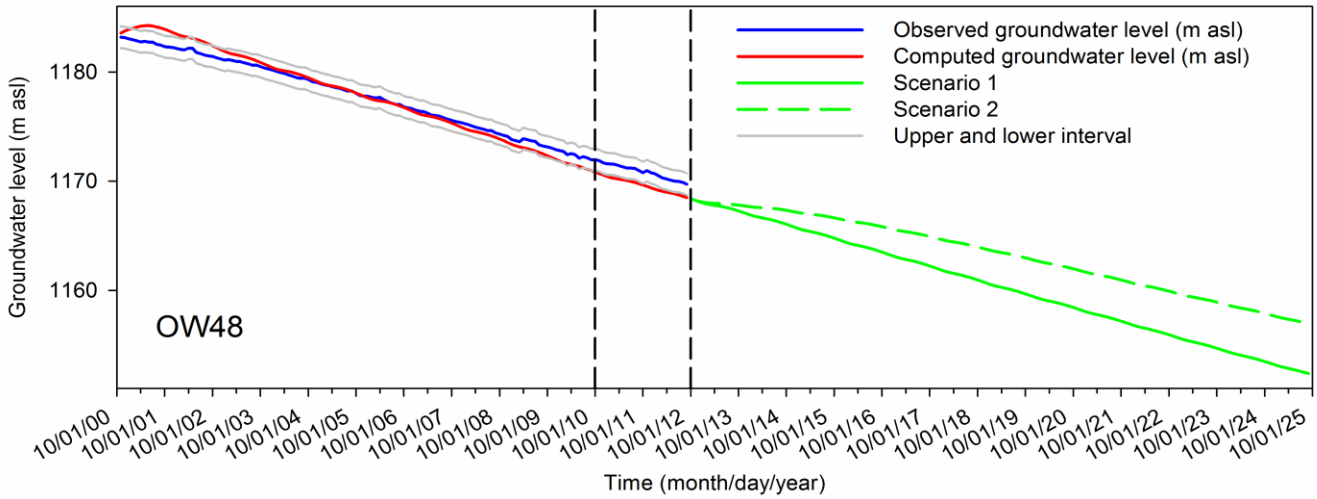

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1140
1130
1120
OW43
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

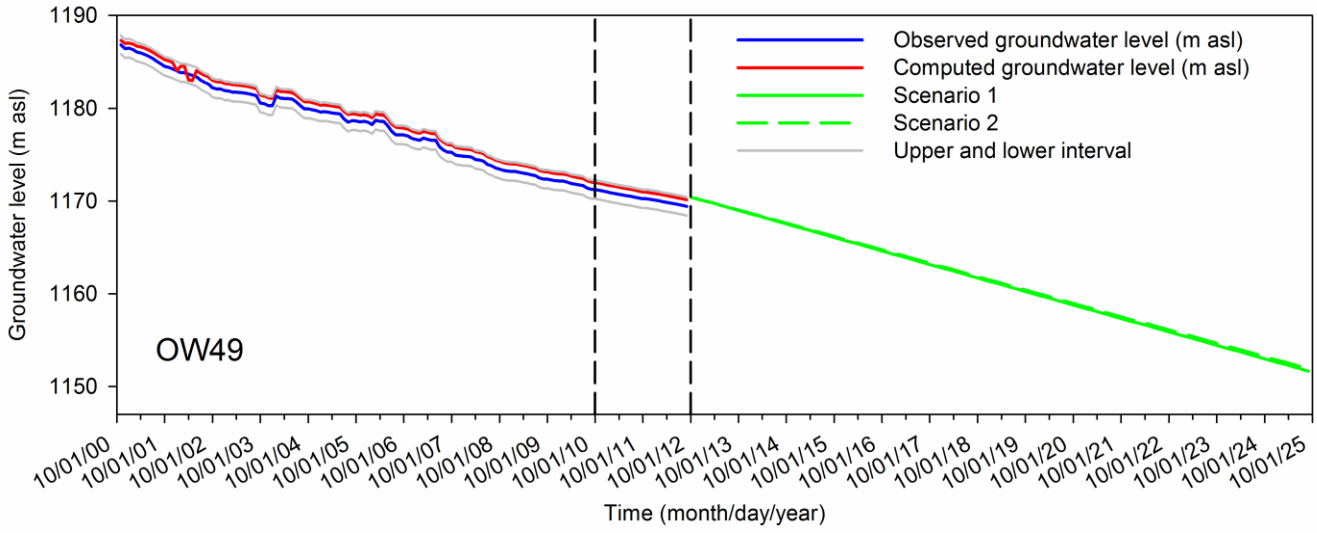

Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1140
1135
1130
1125
1120
1115
OW44
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)


Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1150
1140
1130
1120
1110
OW45
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)


Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1160
1150
1140
1130
1120
1110
1100
OW46
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)


Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1160
1155
1150
1145
1140
1135
OW47
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)


Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1180
1170
1160
OW48
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)


Observed groundwater level (m asl)
Computed groundwater level (m asl)
Scenario 1
Scenario 2
Upper and lower interval
Groundwater level (m asl)
1190
1180
1170
1160
1150
OW49
10/01/00
10/01/01
10/01/02
10/01/03
10/01/04
10/01/05
10/01/06
10/01/07
10/01/08
10/01/09
10/01/10
10/01/11
10/01/12
10/01/13
10/01/14
10/01/15
10/01/16
10/01/17
10/01/18
10/01/19
10/01/20
10/01/21
10/01/22
10/01/23
10/01/24
10/01/25
Time (month/day/year)

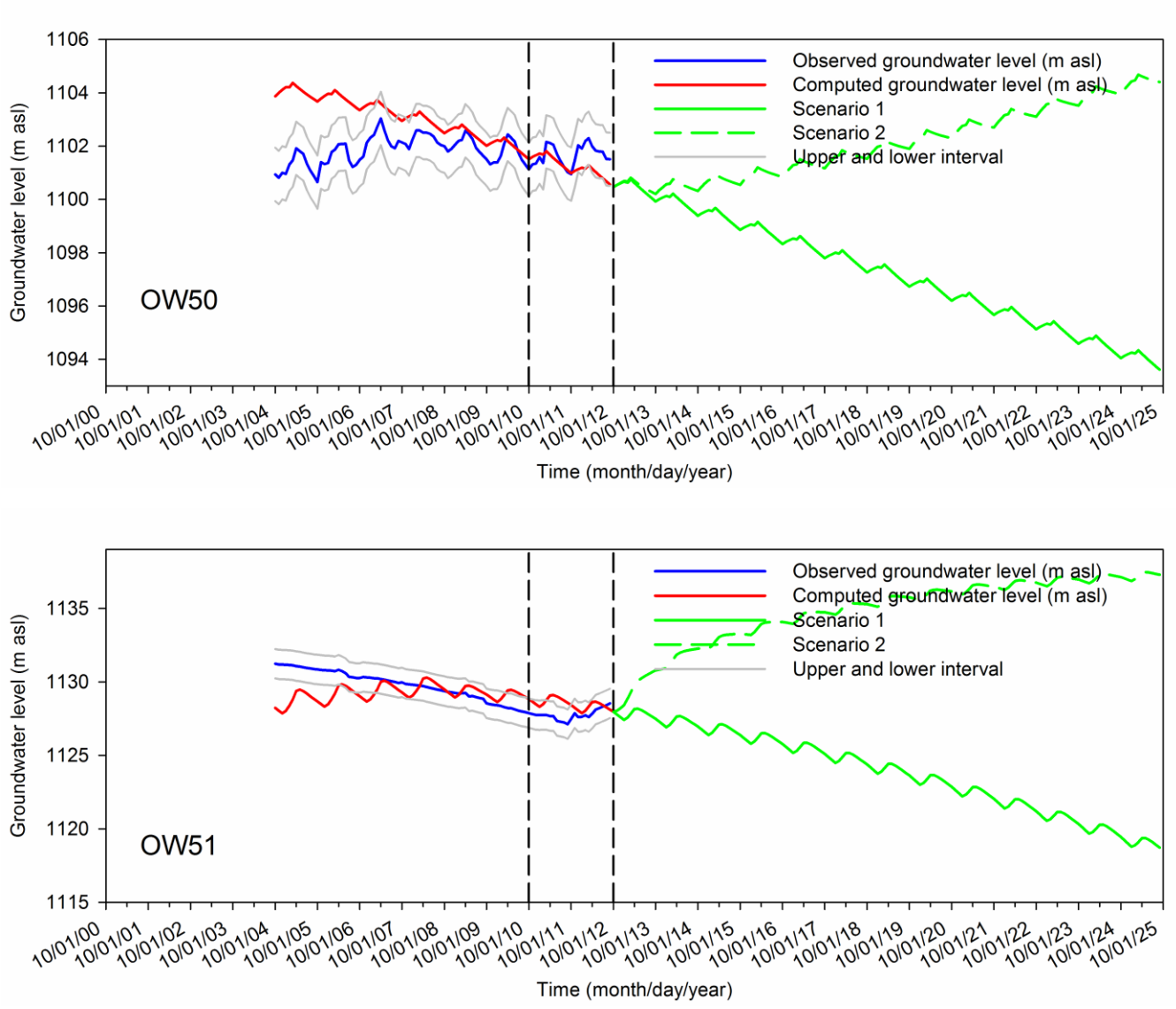


Fig. S8 Plots of observed and computed groundwater level during calibration (Oct. 2000 to Sep. 2010), validation (Oct. 2010 to Sep. 2012) and prediction (Oct. 2012 to Sep. 2025) periods.